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Yu et al.

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(54) **SEMICONDUCTOR DEVICE**

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H10B 12/00 (2023.01)

(52) **U.S. Cl.**
CPC **H10B 12/315** (2023.02); **H10B 12/0335**
(2023.02); **H10B 12/482** (2023.02)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Property Office for Taiwanese Patent Application No. 111127779.

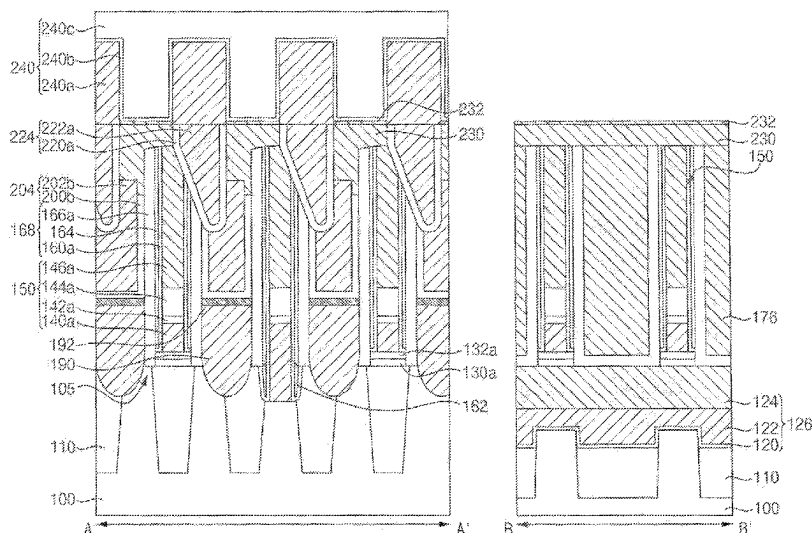
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(57) **ABSTRACT**

A semiconductor device may include bit line structures on a substrate, a contact plug structure on the substrate between the bit line structures, and a capacitor electrically connected to the contact plug structure. The contact plug structure may include a first contact plug, a second contact plug, and a third contact plug sequentially stacked. An upper surface of the second contact plug includes an upper recess. The third contact plug may fill the upper recess, and may protrude above the upper recess. An upper surface of the third contact plug may be higher than a top surface of the bit line structures.

20 Claims, 26 Drawing Sheets



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FIG. 1

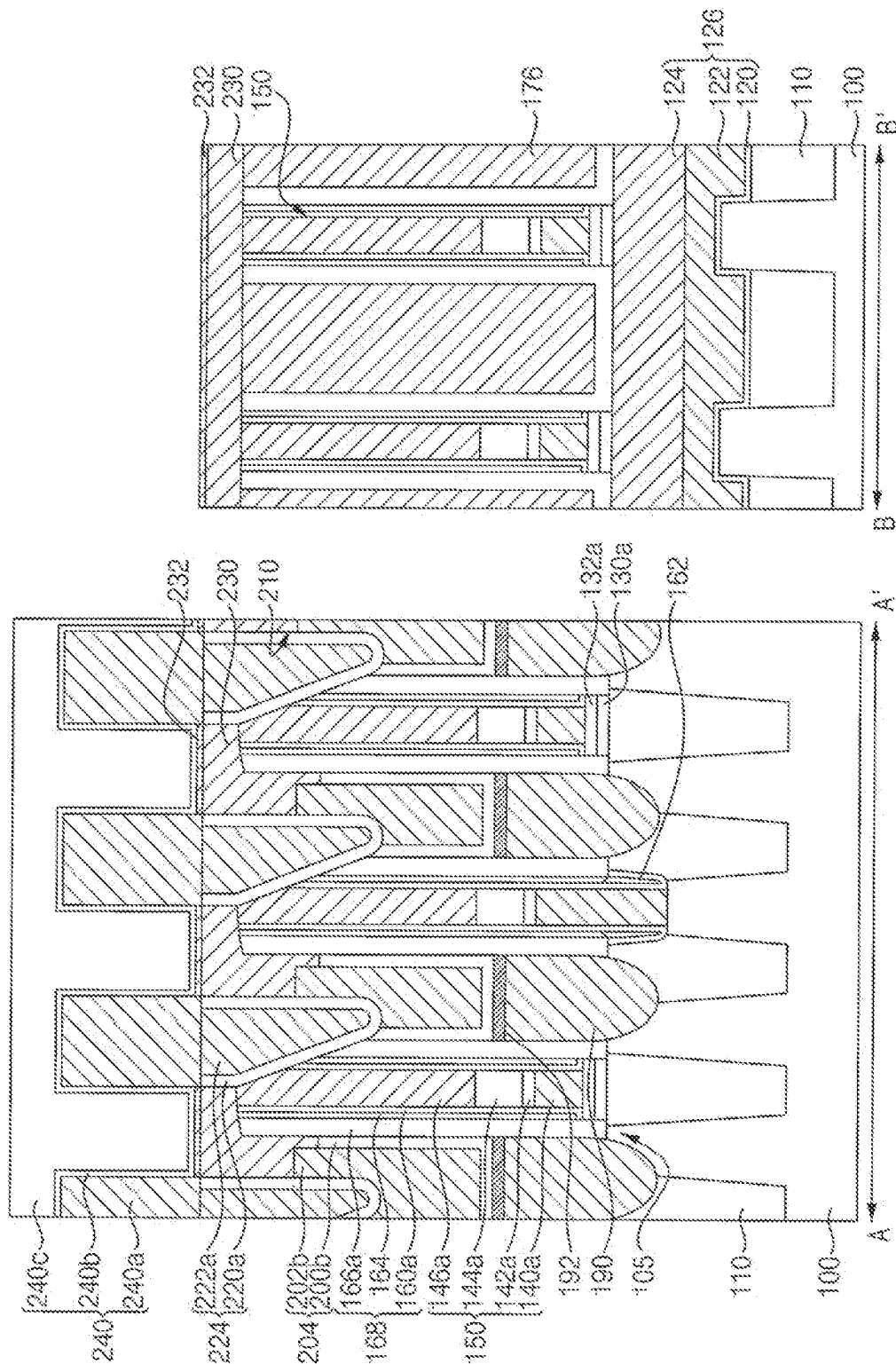


FIG. 2

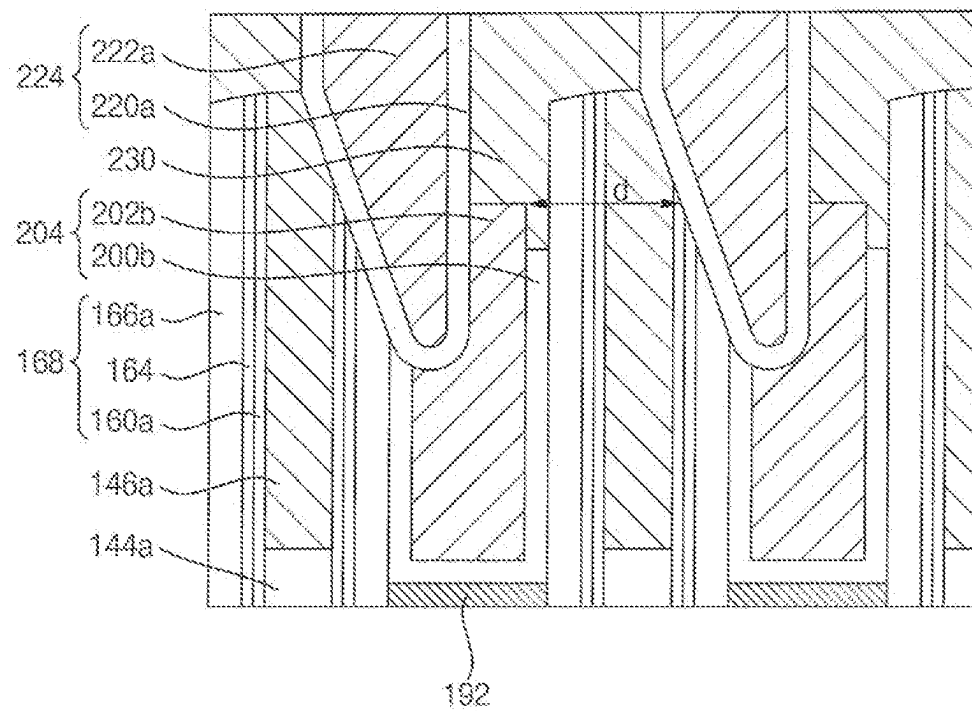


FIG. 3

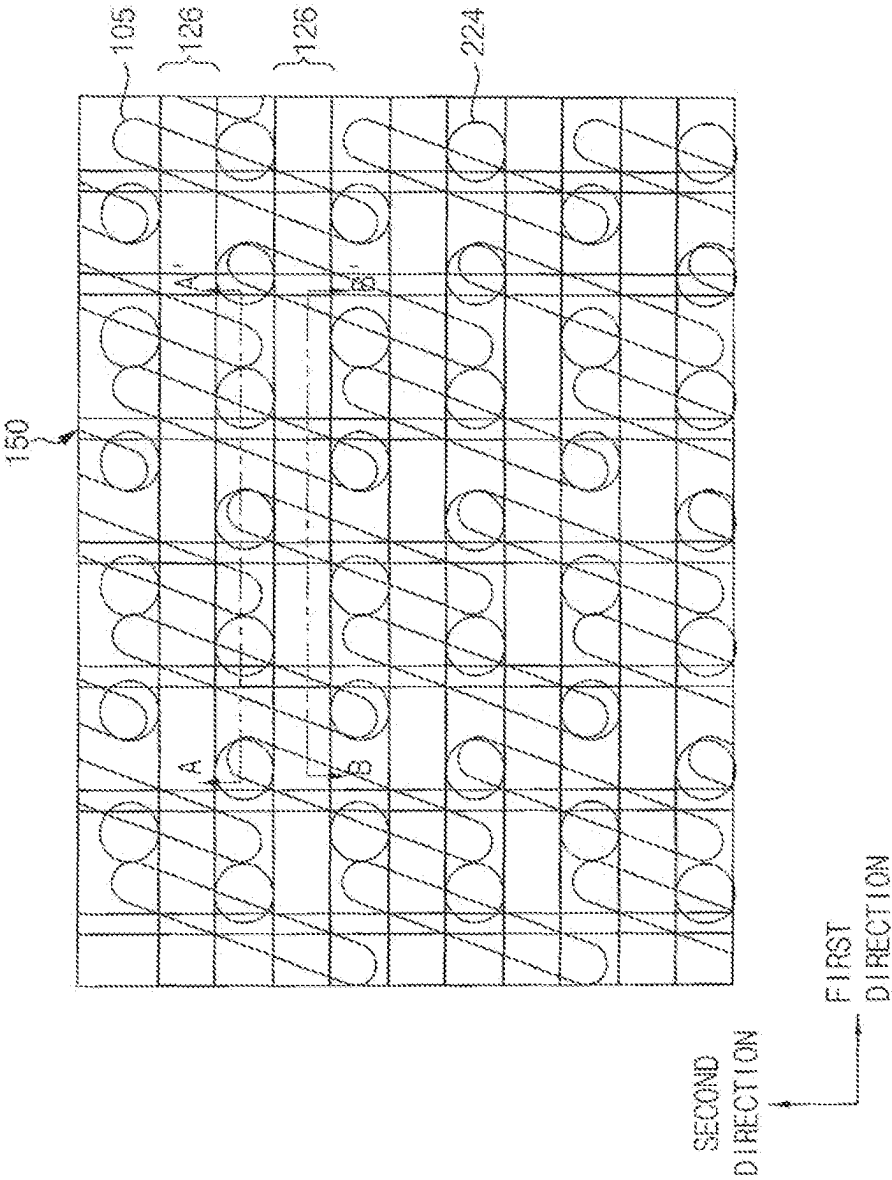


FIG. 4

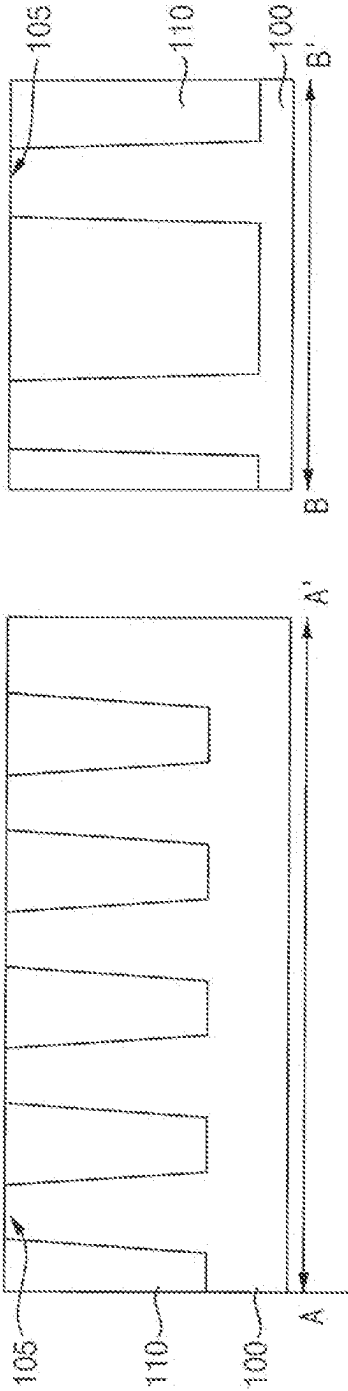
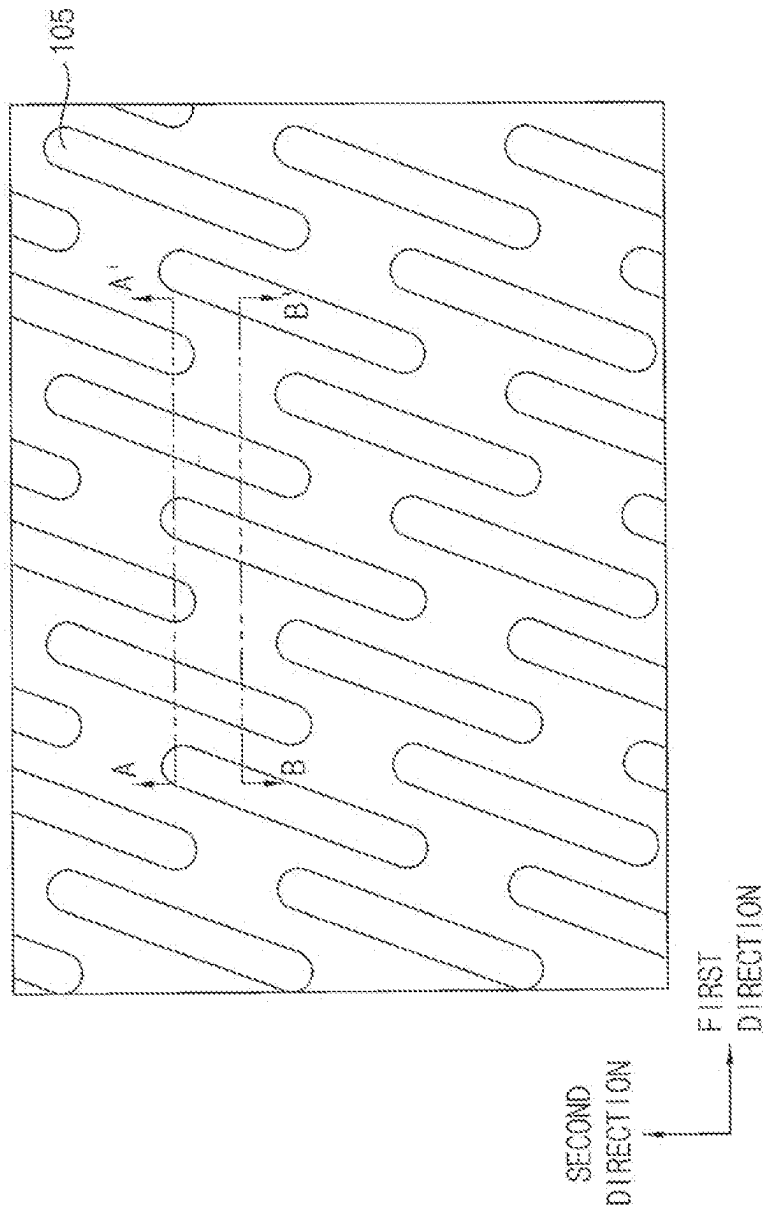


FIG. 5



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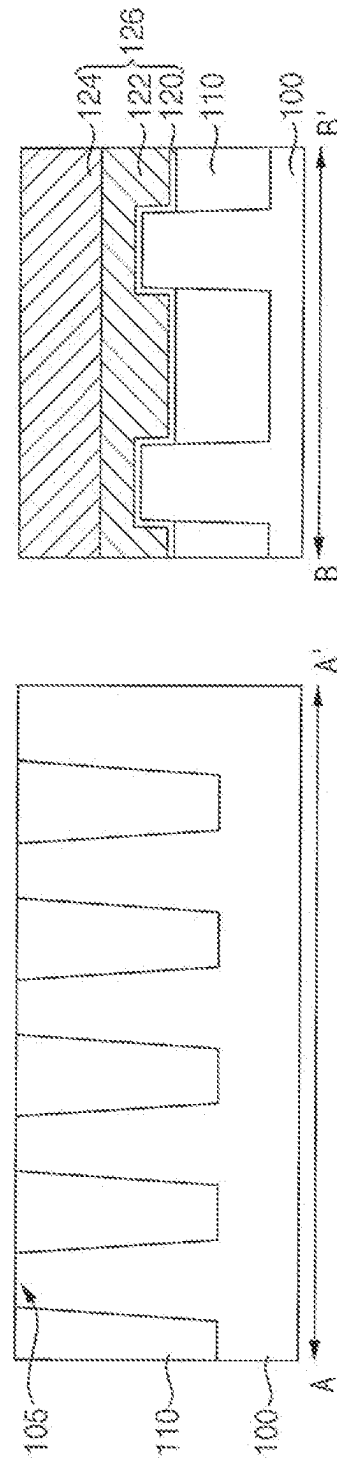


FIG. 7

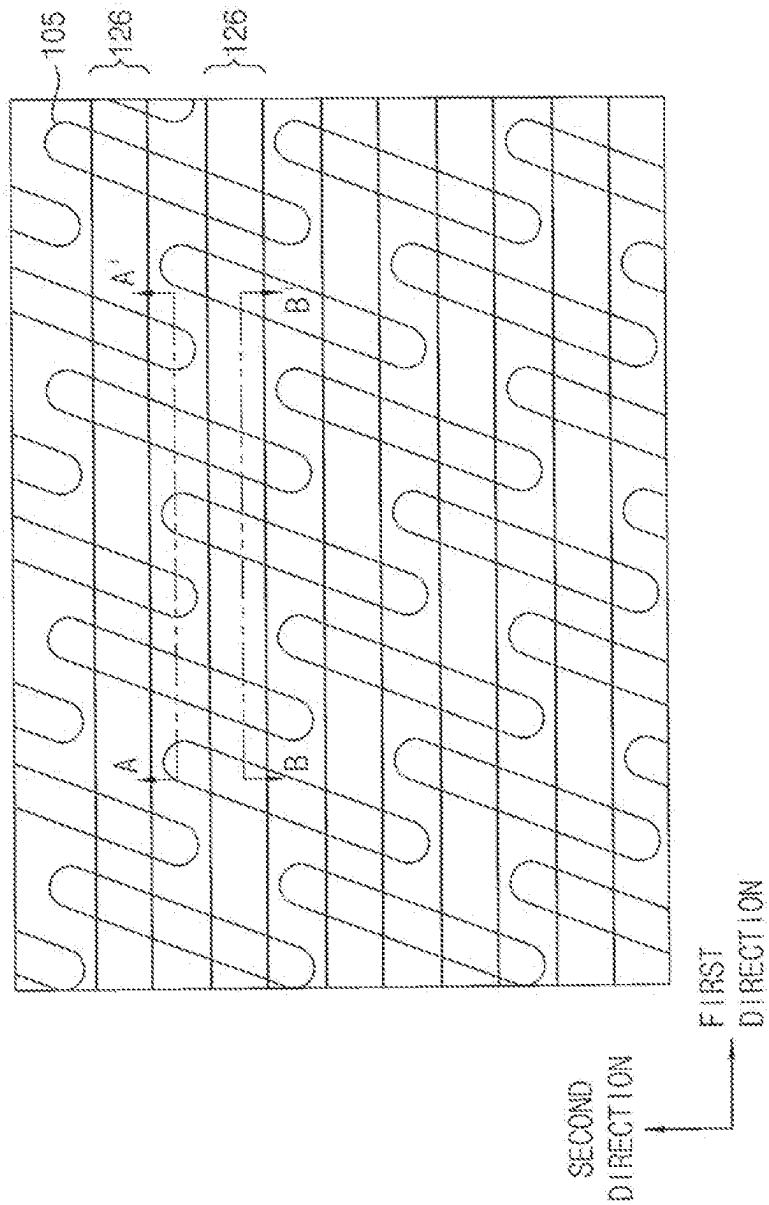


FIG. 8

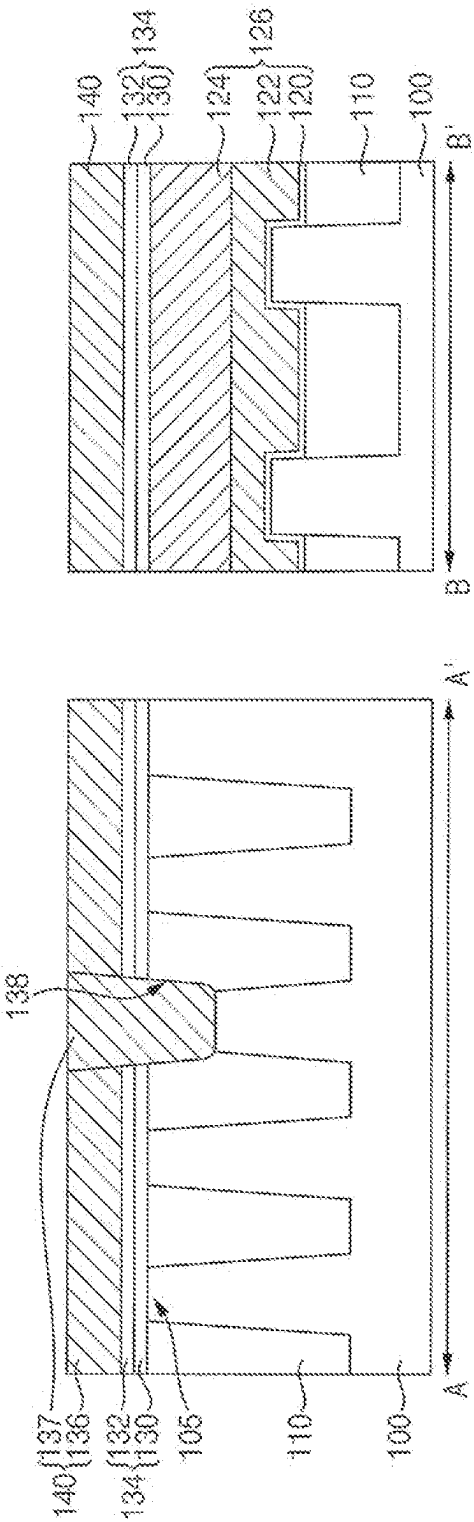


FIG. 9

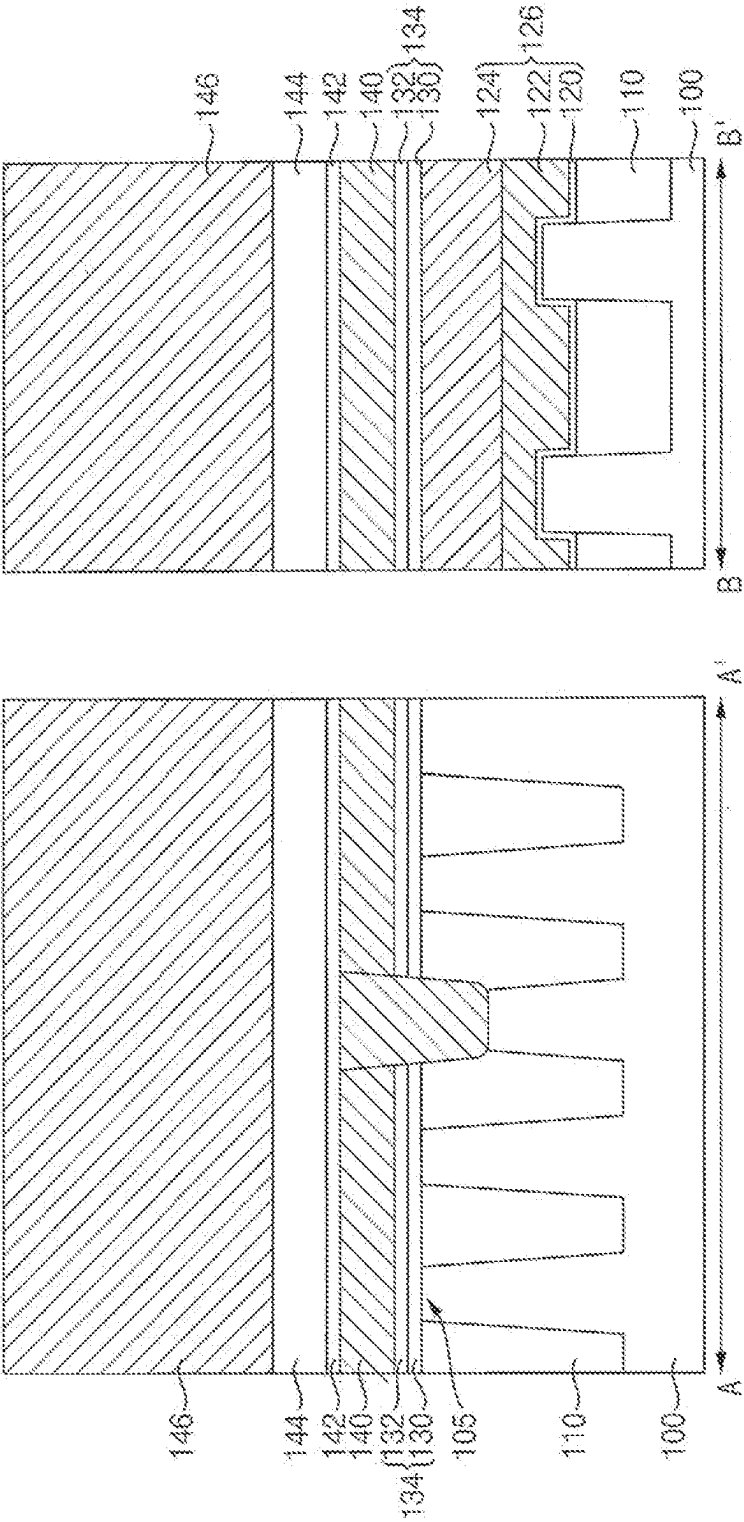


FIG. 10

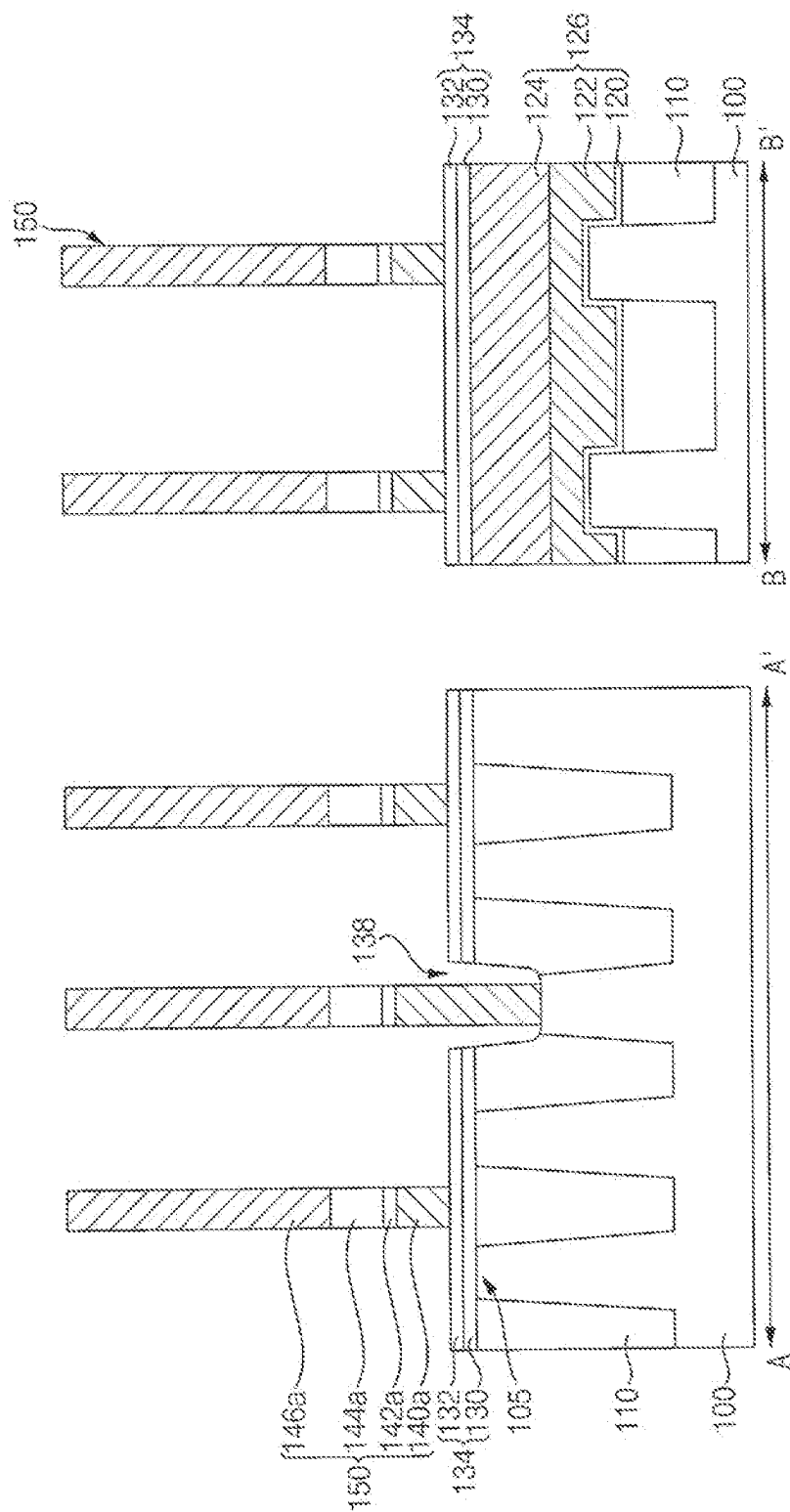
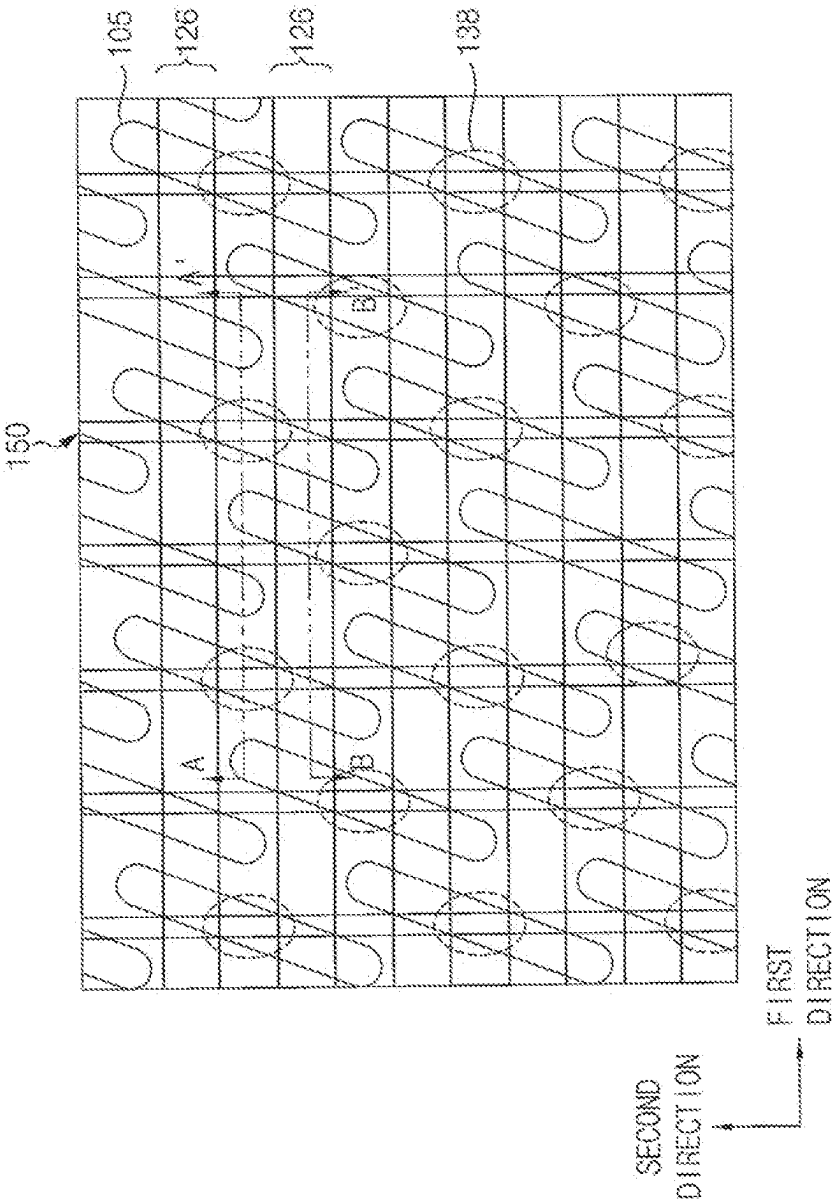


FIG. 11



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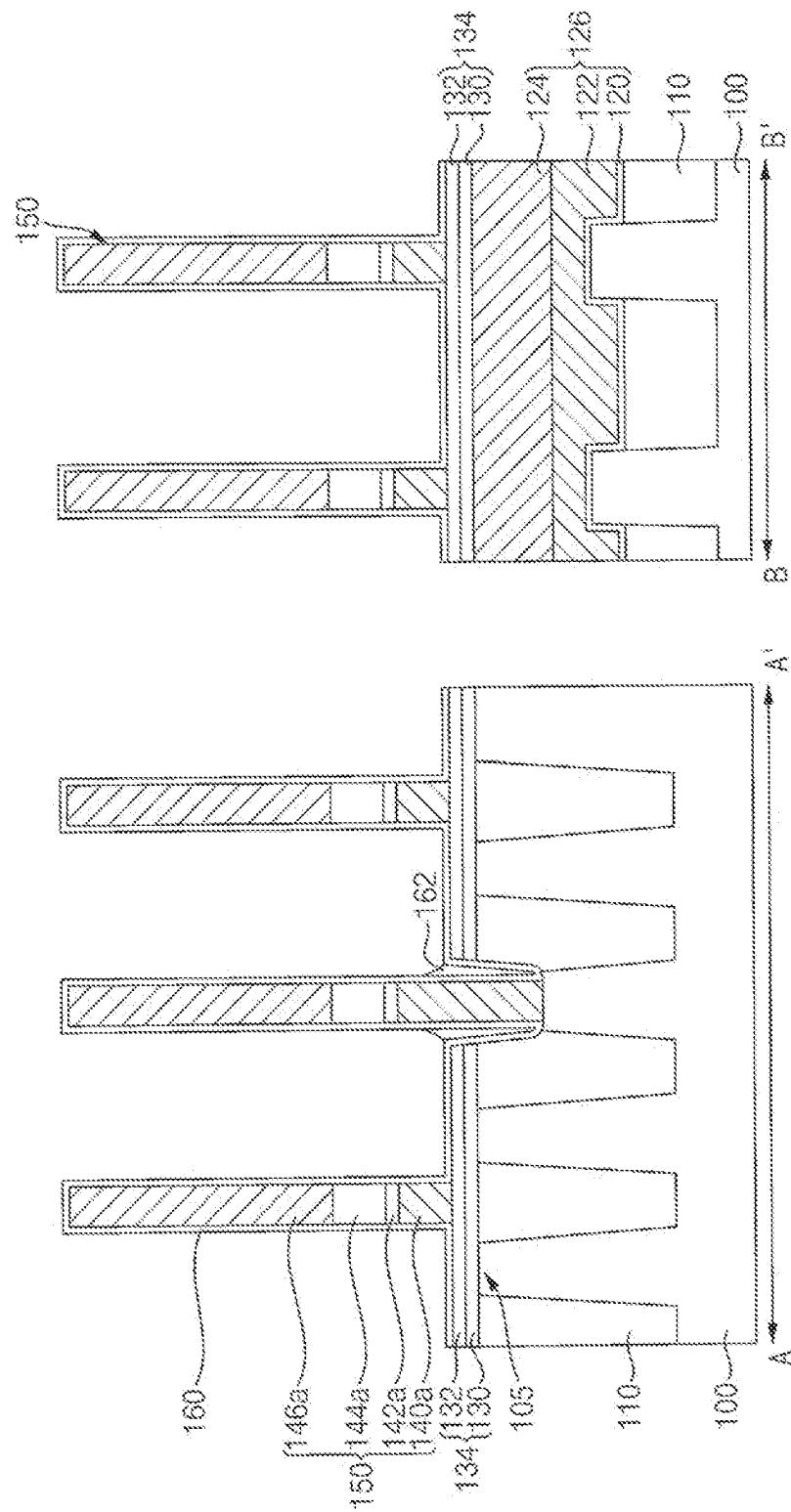


FIG. 13

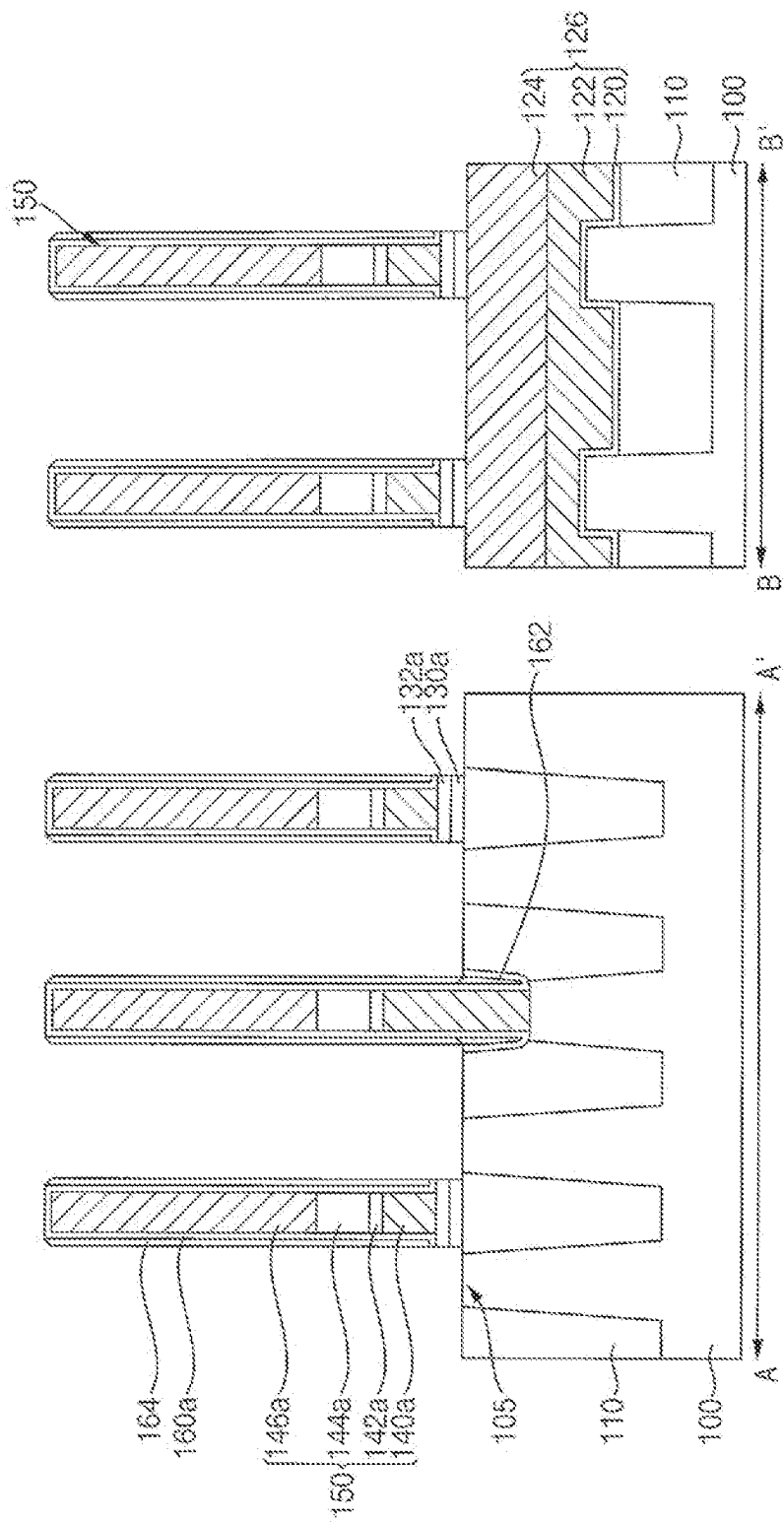


FIG. 14

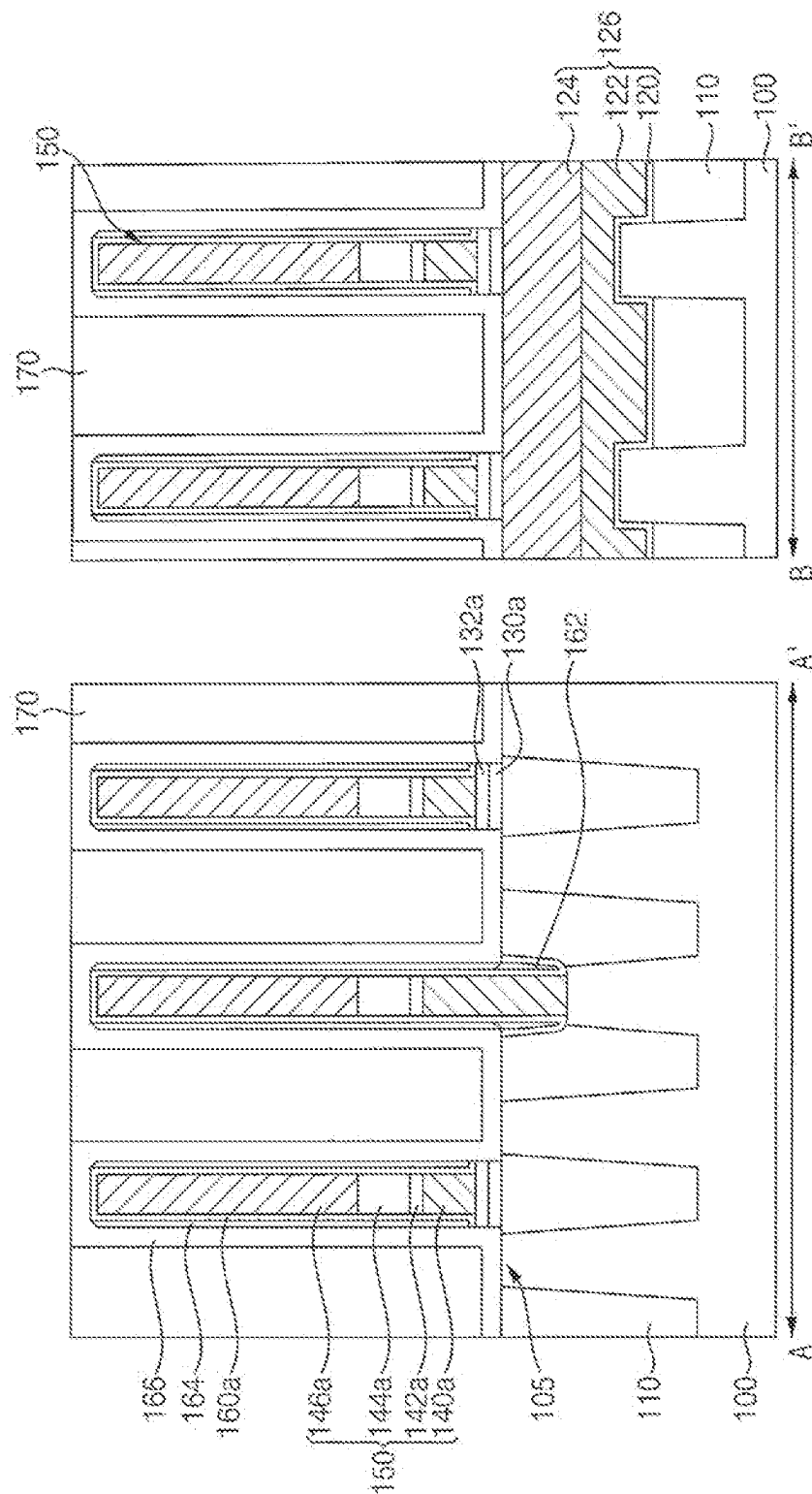


FIG. 15

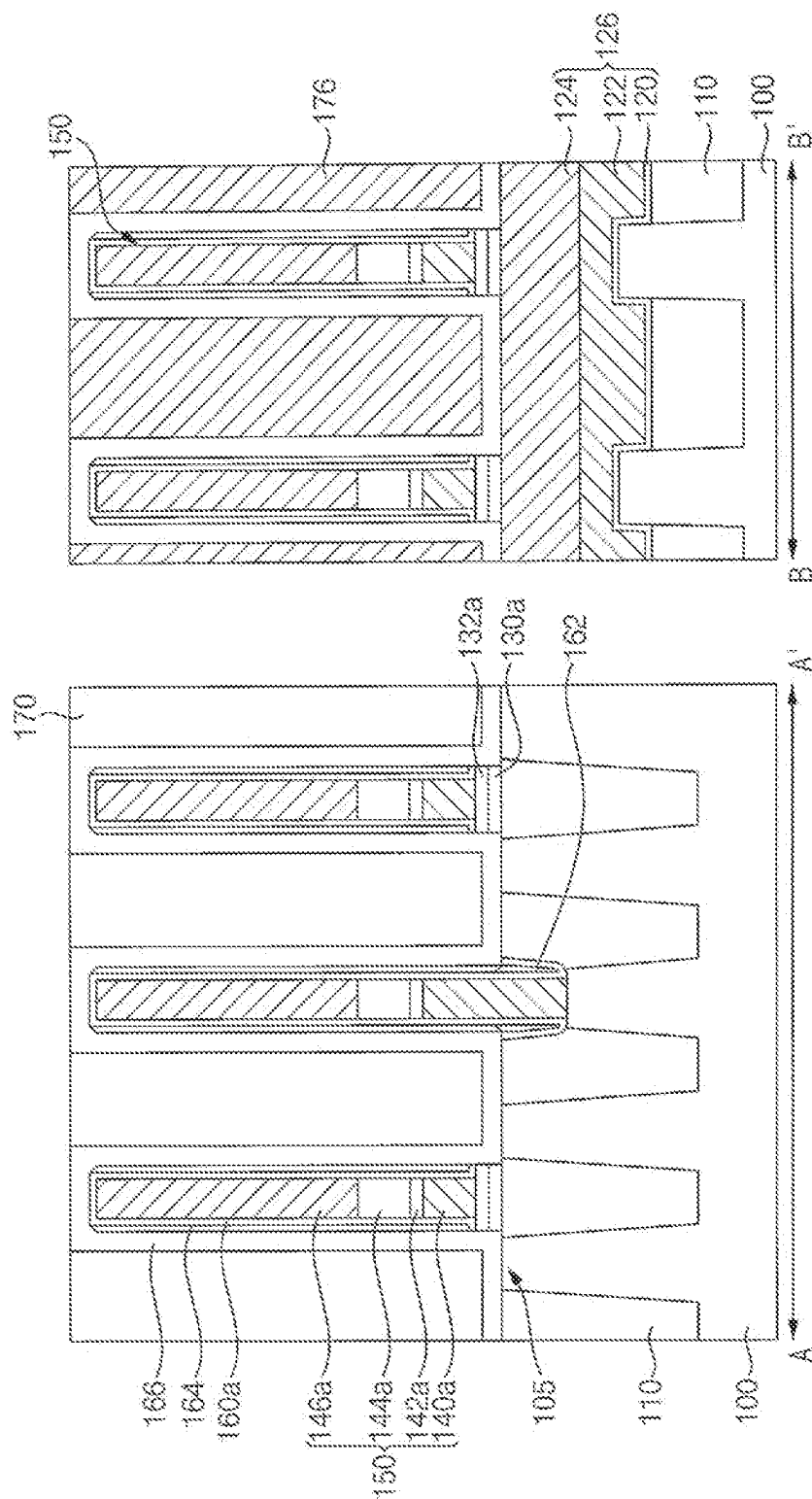


FIG. 16

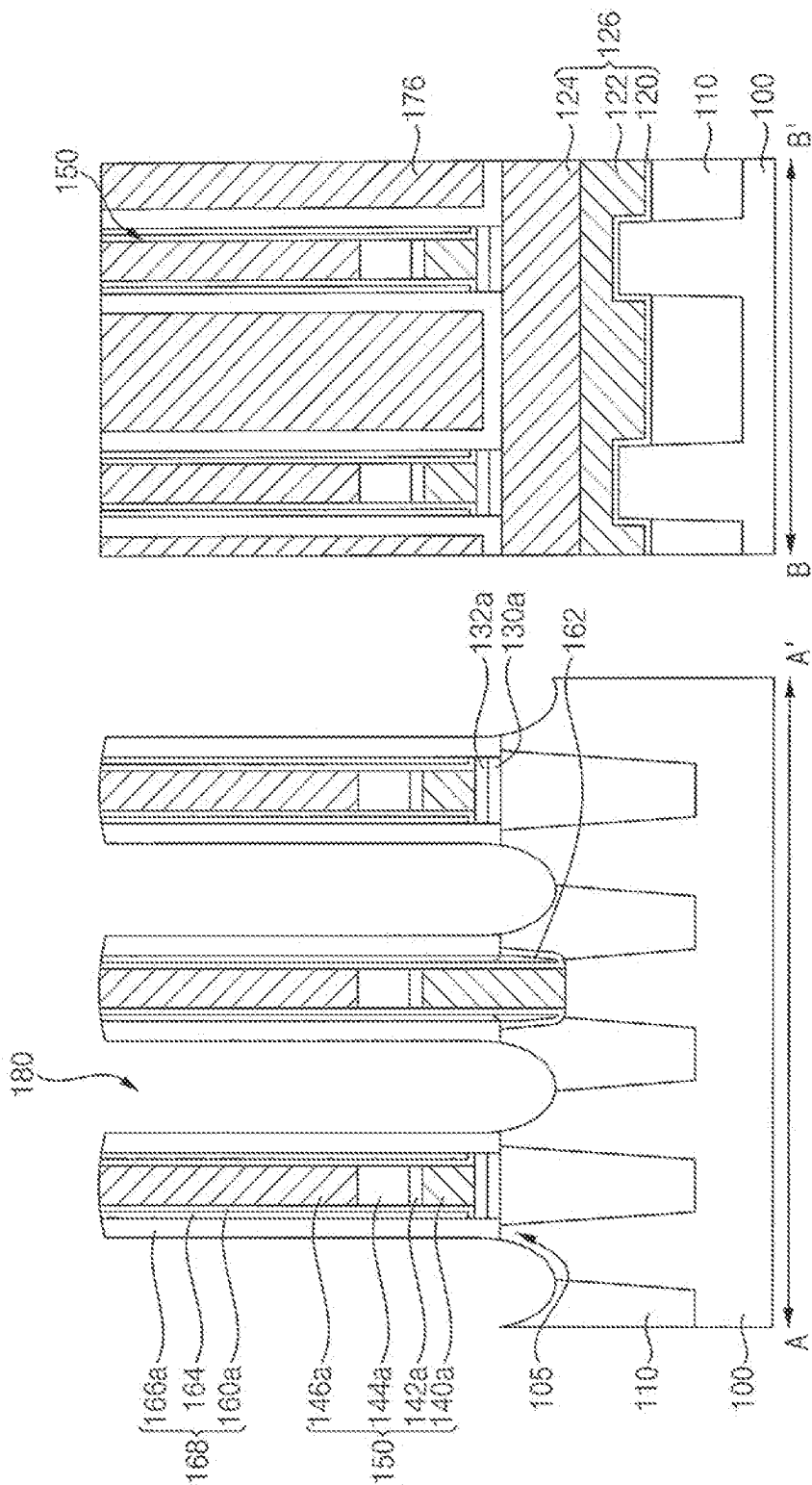


FIG. 18

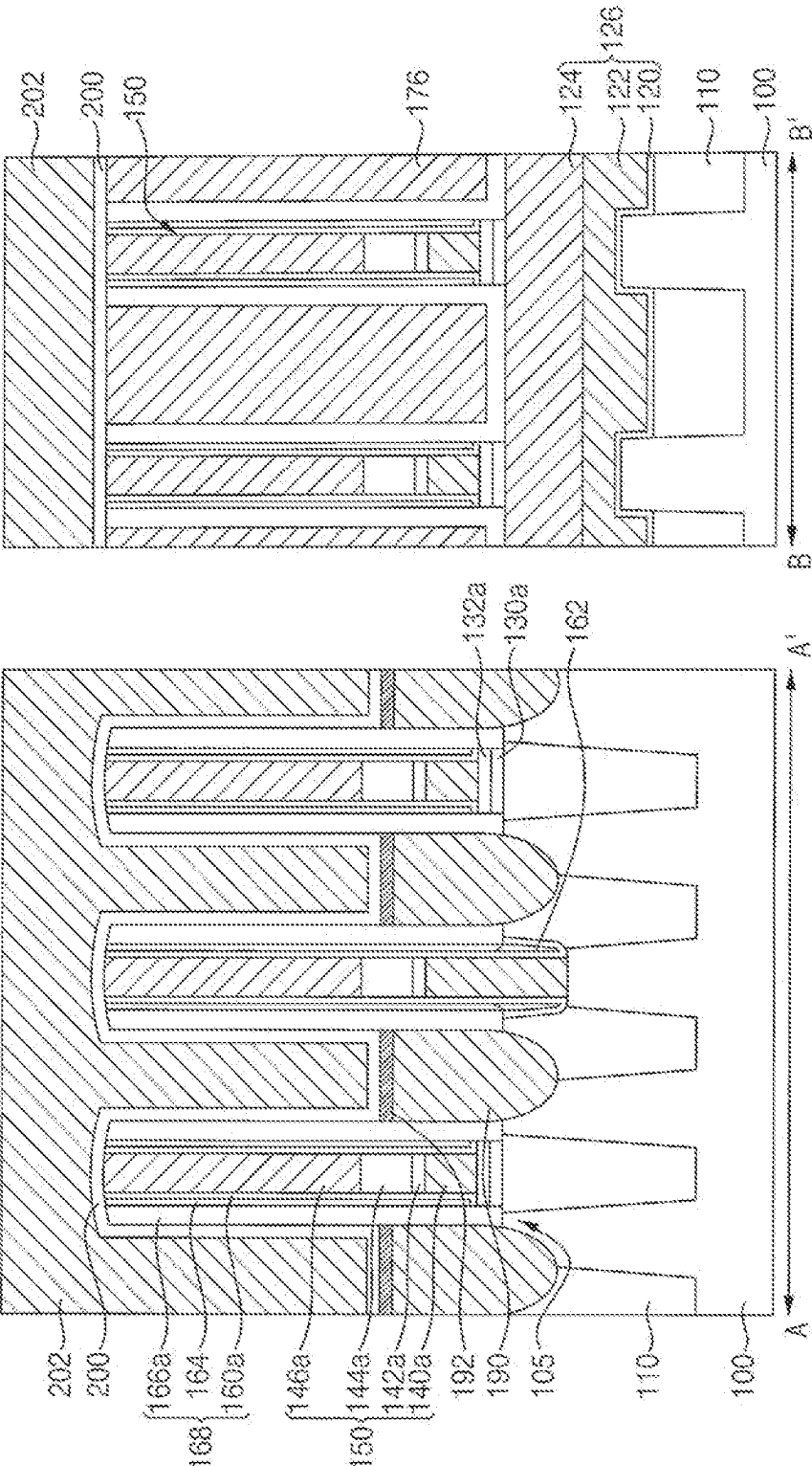


FIG. 19

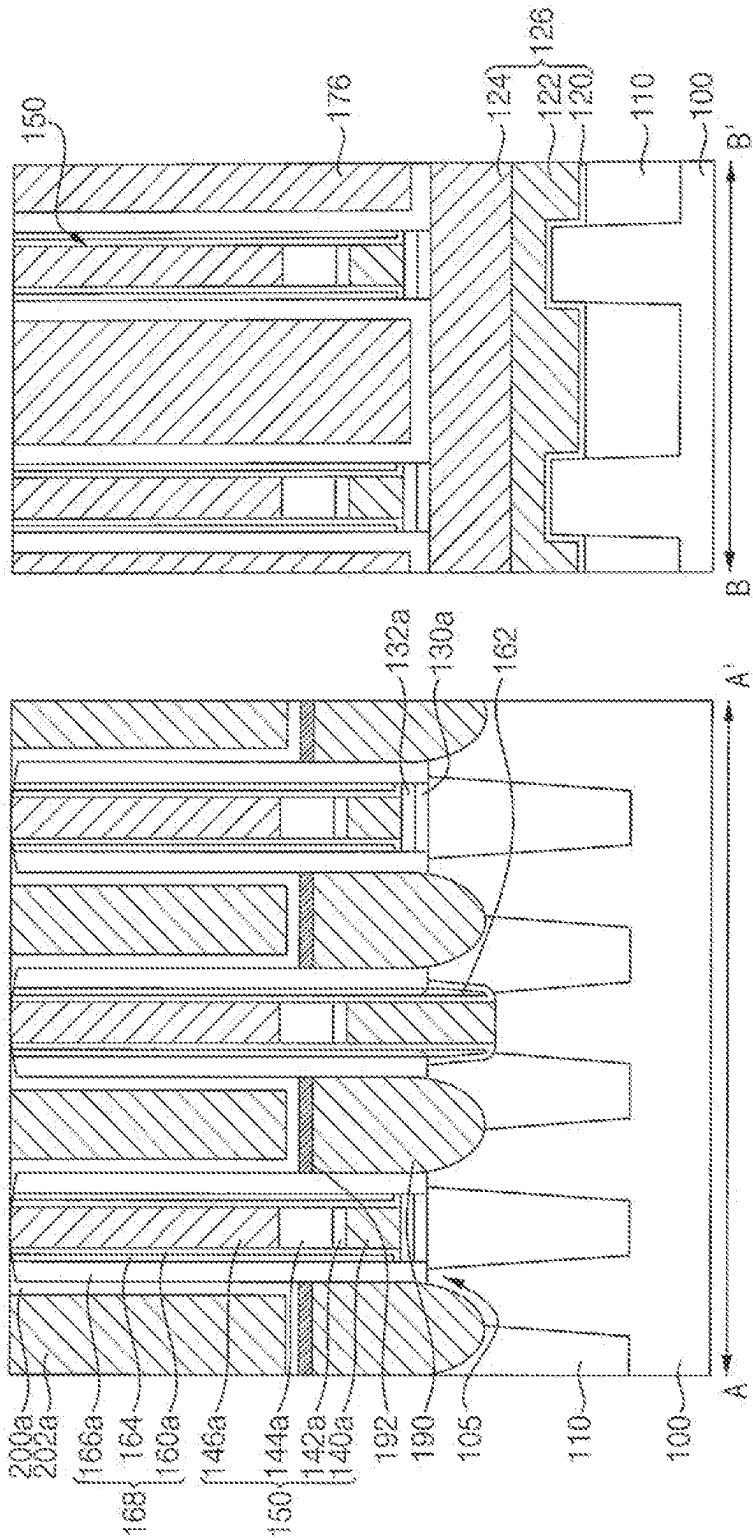


FIG. 20

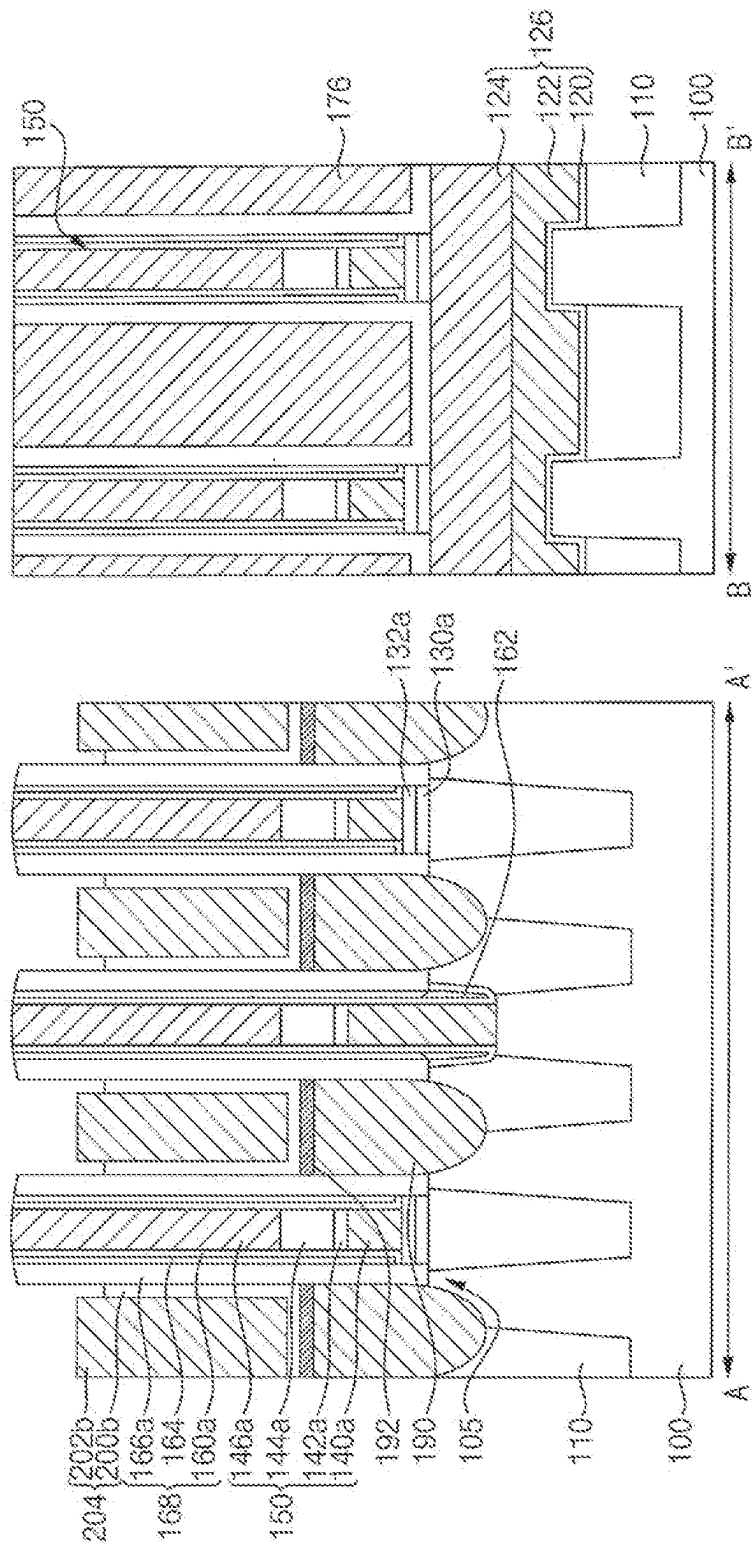


FIG. 21

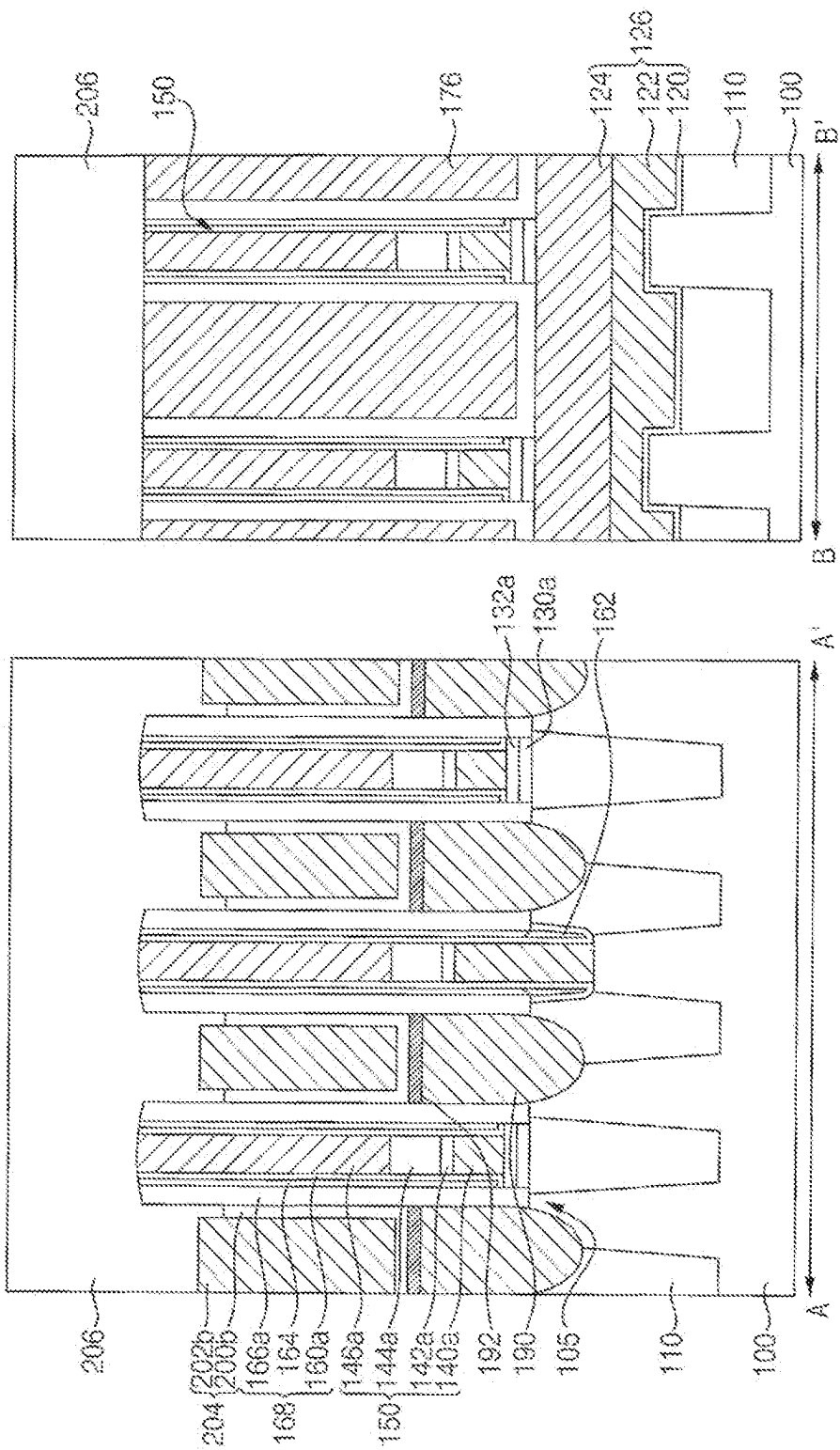


FIG. 22

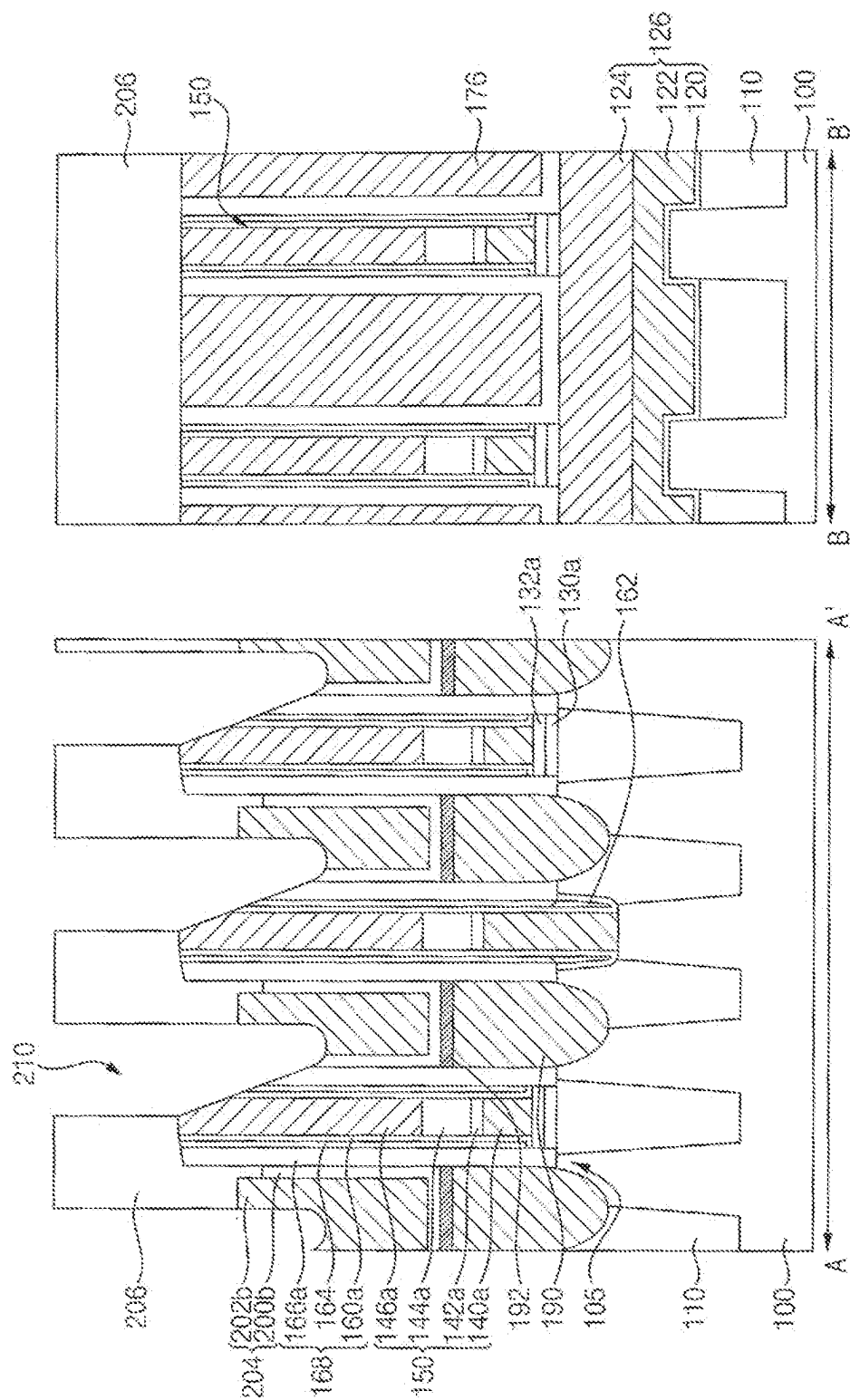


FIG. 23

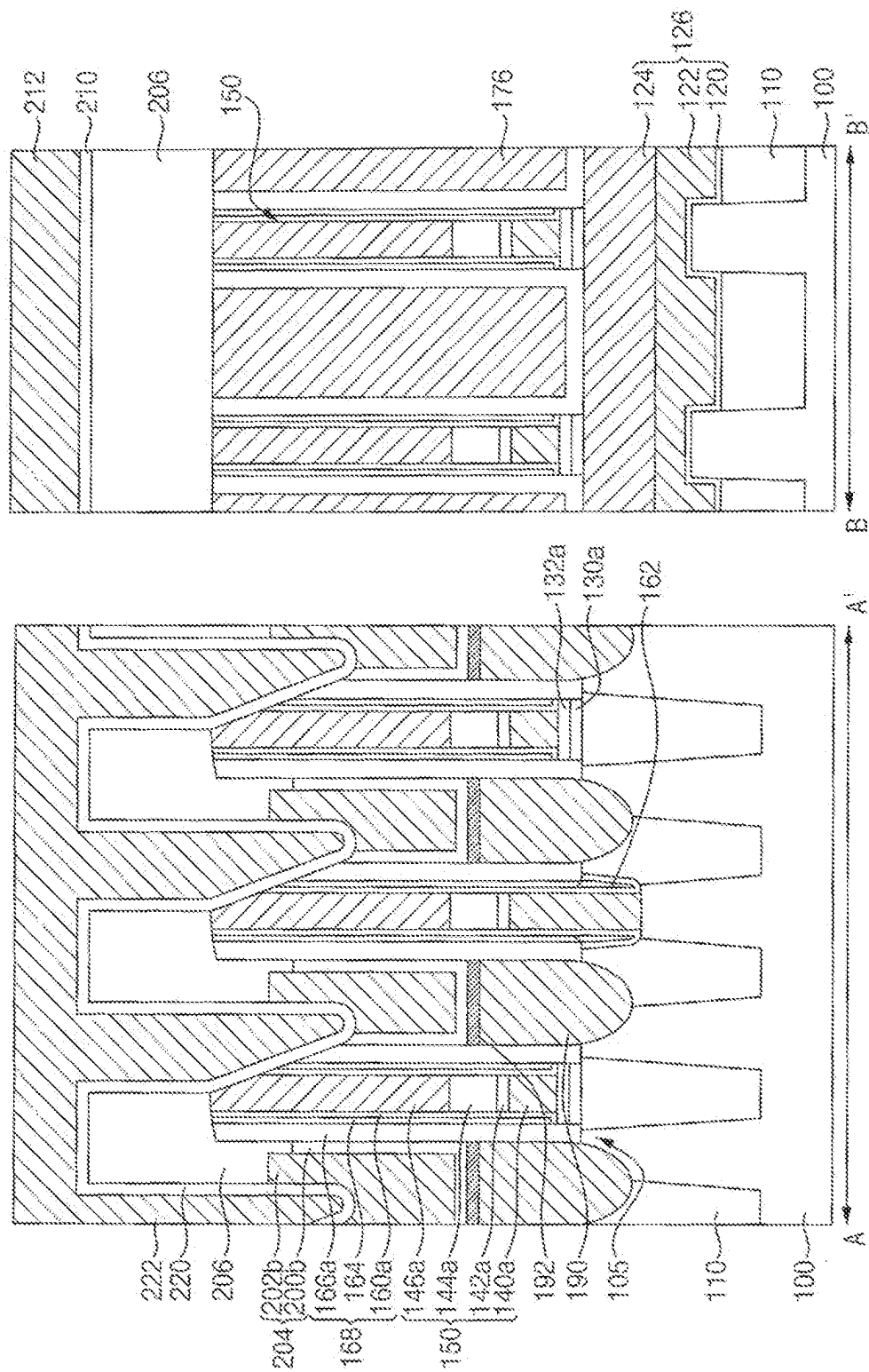
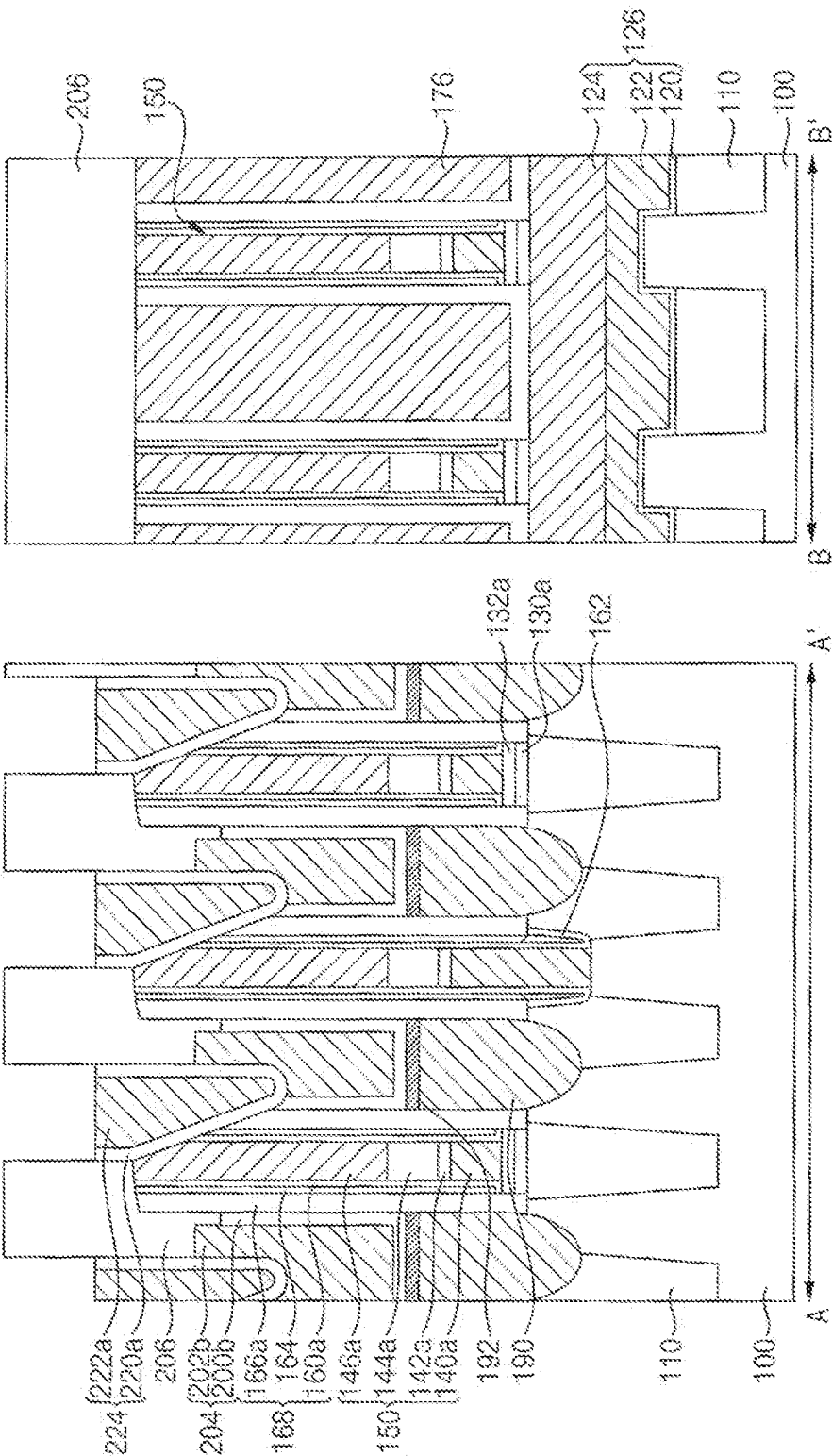


FIG. 24



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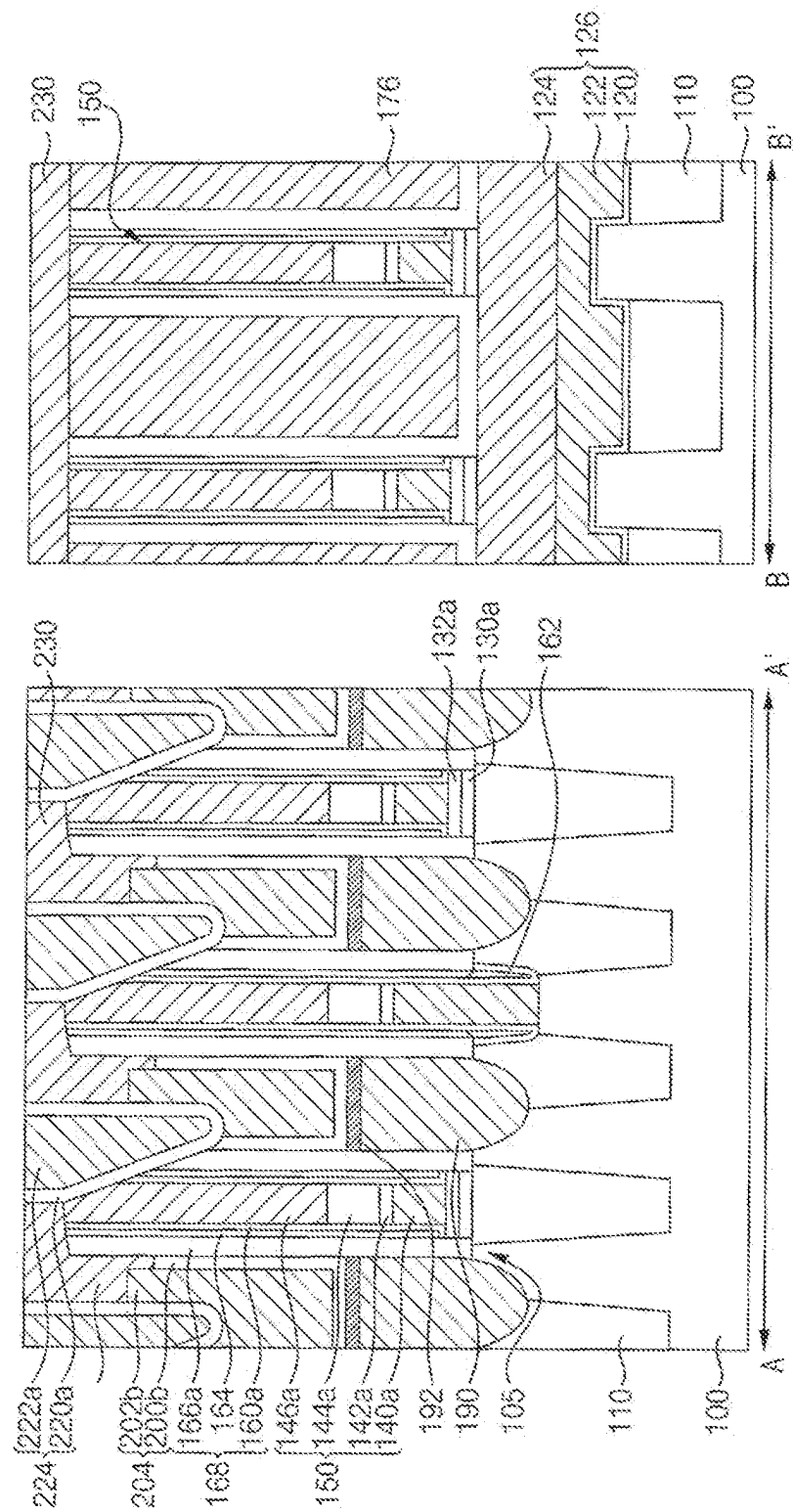
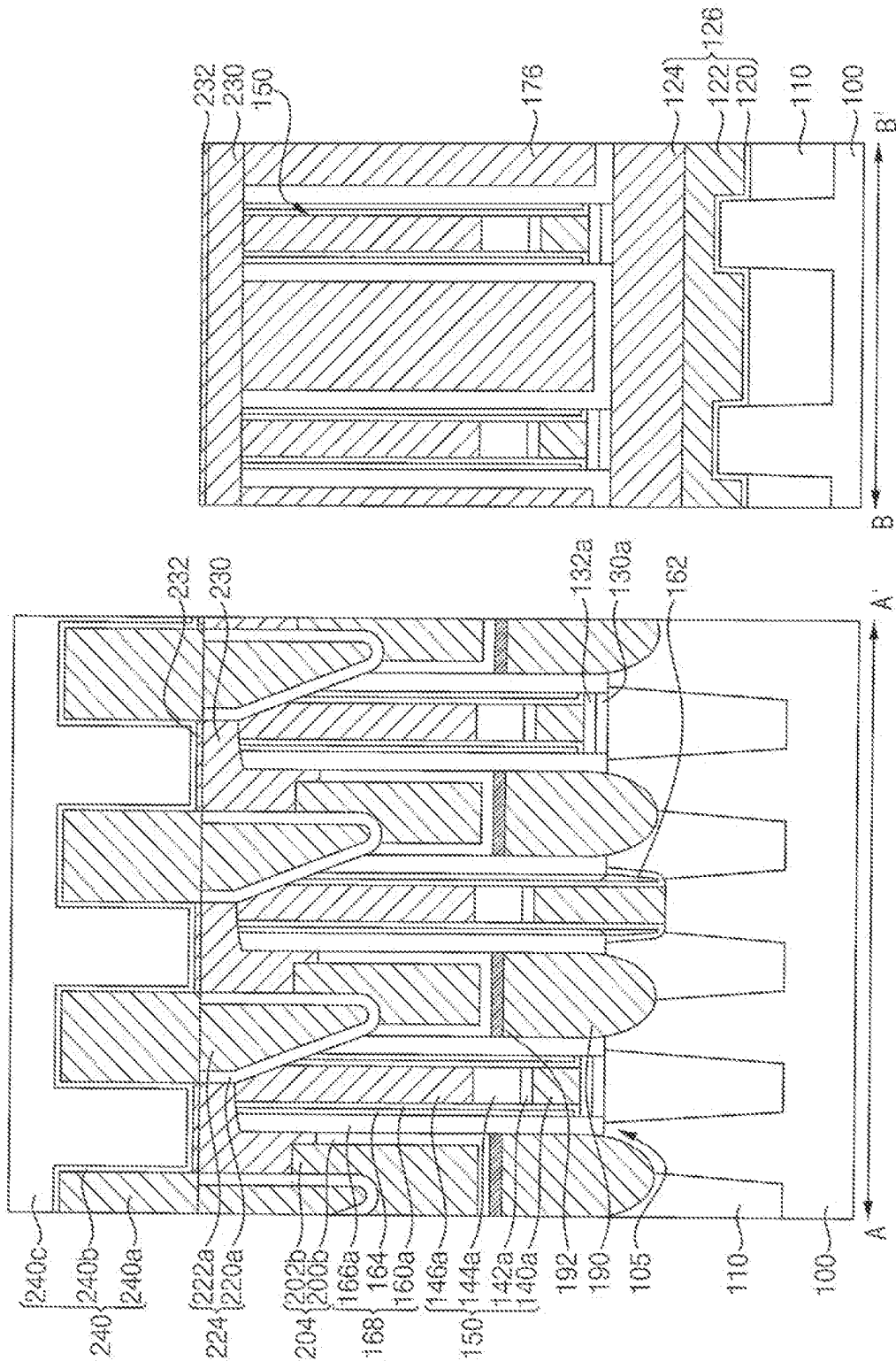


FIG. 26



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SEMICONDUCTOR DEVICE**CROSS-REFERENCE TO RELATED APPLICATION**

This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0139079, filed on Oct. 19, 2021, in the Korean Intellectual Property Office (KIPO), the contents of which are incorporated by reference herein in their entirety.

BACKGROUND**1. Field**

Embodiments relate to a semiconductor device. More particularly, embodiments relate to a DRAM device.

2. Description of the Related Art

As a dynamic random access memory (DRAM) device is highly integrated, a contact plug structure for electrically connecting a lower impurity region and an upper capacitor may have a high height. Therefore, an upper portion of the contact plug structure may be broken, or a bridge failure between adjacent contact plug structures may occur. Thus, an operation failure of the DRAM device may occur.

SUMMARY

Example embodiments provide a semiconductor device having good characteristics.

Example embodiments provide a method for manufacturing a semiconductor device having good characteristics.

According to example embodiments, there is provided a semiconductor device. The semiconductor device may include bit line structures on a substrate, a contact plug structure on the substrate between the bit line structures, and a capacitor electrically connected to the contact plug structure. The contact plug structure may include a first contact plug, a second contact plug, and a third contact plug sequentially stacked. An upper surface of the second contact plug includes an upper recess. The third contact plug may fill the upper recess, and may protrude above the upper recess. An upper surface of the third contact plug may be higher than a top surface of the bit line structures.

According to example embodiments, there is provided a semiconductor device. The semiconductor device may include bit line structures on a substrate, a contact plug structure on the substrate between the bit line structures, and a capacitor electrically connected to the contact plug structure. The contact plug structure may include a first contact plug including polysilicon, a second contact plug including a metal on the first contact plug, and a third contact plug including a metal on the second contact plug sequentially stacked. A top surface of the second contact plug may be lower than a top surface of the bit line structures. An upper surface of the third contact plug may be higher than the top surface of the bit line structures. A lowermost of the third contact plug may be lower than the top surface of the second contact plug. A first sidewall of the third contact plug may contact a sidewall of a first bit line structure of the bit line structures.

According to example embodiments, there is provided a semiconductor device. The semiconductor device may include a substrate including an active pattern defined by an isolation pattern, gate structures buried in the active pattern

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and the isolation pattern of the substrate, bit line structures on the substrate, a contact plug structure on the active pattern between the bit line structures, and a capacitor electrically connected to the contact plug structure. The contact plug structure may include a first contact plug including polysilicon, a second contact plug including a metal on the first contact plug, and a third contact plug including a metal on the second contact plug sequentially stacked. The third contact plug may contact a sidewall of a first bit line structure of the bit line structures and a portion of an upper surface of the second contact plug. Both sidewalls of the third contact plug may have slopes different from each other, in a cross-sectional view.

According to example embodiments, there is provided a method for manufacturing a semiconductor device. In the method, bit line structures may be formed on a substrate. A first contact plug may be formed on the substrate between the bit line structures. A second contact plug may be formed on the first contact plug. A sacrificial insulation layer may cover the bit line structures and the first and second contact plugs. A portion of the sacrificial insulation layer, an upper portion of a first bit line structure of the bit line structures, and an upper portion of the second contact plug may be etched to form an opening. A third contact plug may be formed in the opening. The third contact plug may contact the second contact plug. A capacitor may be formed on the third contact plug. A top surface of the second contact plug may be lower than a top surface of the bit line structures. An upper surface of the third contact plug may be higher than the top surface of the bit line structures.

The semiconductor device in accordance with example embodiments may include the contact plug structure including the first contact plug, the metal silicide pattern, the second contact plug, and the third contact plug sequentially stacked. The contact plug structure may be electrically connected with the impurity region and the capacitor. The third contact plug may contact an upper portion of the second contact plug, the capping pattern in an upper portion of the bit line structure and the spacer structure. A portion of the third contact plug contacting the capping pattern and the spacer structure may have a slope such that a width may be gradually decreased downward of the third contact plug. Thus, an upper width of the third contact plug may be greater than a lower width of the third contact plug. As described above, the upper width of the third contact plug may increase, so that a defect in which an upper portion of the third contact plug is broken may be decreased. In addition, a bottom of the third contact plug may contact only a portion of an upper surface of the second contact plug, so that a gap between upper portions of adjacent contact plug structures may be increased. Therefore, bridge defects between adjacent contact plug structures may be decreased.

Further, in the method of manufacturing the semiconductor device, the third contact plug may be formed by performing a damascene process instead of an embossed patterning process. Accordingly, the upper width of the third contact plug may be sufficiently wide, so that defects due to decreasing of the upper width of the third contact plug may be decreased.

BRIEF DESCRIPTION OF THE DRAWINGS

Example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which like numerals refer to like elements throughout. FIGS. 1 to 26 represent non-limiting, example embodiments as described herein.

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FIGS. 1 to 3 are cross-sectional views and a plan view illustrating semiconductor devices in accordance with example embodiments; and

FIGS. 4 to 26 are cross-sectional views and plan views illustrating a method of manufacturing a semiconductor device in accordance with example embodiments.

DESCRIPTION OF EMBODIMENTS

FIGS. 1 and 2 are cross-sectional views illustrating semiconductor devices in accordance with example embodiments. FIG. 2 is a plan view illustrating semiconductor devices in accordance with example embodiments.

The cross-sectional views of FIG. 1 include cross-sectional views taken along lines A-A' and B-B' in the plan view of FIG. 3. FIG. 2 is an enlarged cross-sectional view of a portion of a contact plug structure in the semiconductor device of FIG. 1.

Referring to FIGS. 1 and 2, the semiconductor device may include a gate structure 126 buried in a substrate 100, and a bit line structure 150, a spacer structure 168, a contact plug structure, and a capacitor 240 formed on the substrate 100. In addition, the semiconductor device may further include first and second insulation patterns 130a and 132a, a lower insulation pattern 162, a fence insulation pattern 176, an etch stop layer 232, and a second insulating interlayer 230. As used herein, the term "buried" may refer to structures, patterns, and/or layers that are formed at least partially below a top surface of another structure, pattern, and/or layer. In some embodiments, when a first structure, pattern, and/or layer is "buried" in a second structure, pattern, and/or layer, the second structure, pattern, and/or layer may surround at least a portion of the first structure, pattern, and/or layer. For example, a first structure, pattern, and/or layer first may be considered to be buried when it is at least partially embedded in a second structure, pattern, and/or layer.

The substrate 100 may include silicon, germanium, silicon-germanium, or a group III-V compound such as GaP, GaAs, or GaSb. In some example embodiments, the substrate 100 may be a silicon-on-insulator (SOI) substrate or a germanium-on-insulator (GOI) substrate.

An isolation pattern 110 may be formed in the substrate 100, and an active pattern 105 having sidewalls surrounded by the isolation pattern 110 may be defined at a surface of the substrate 100. The isolation pattern 110 may include, e.g., an oxide such as silicon oxide.

In example embodiments, a plurality of active patterns 105 may be spaced apart from each other in each of a first direction parallel to an upper surface of the substrate 100 and a second direction perpendicular to the first direction. Each of the active patterns 105 may extend in a third direction having an acute angle with each of the first and second directions and parallel to the upper surface of the substrate 100. Impurity regions (not shown) may be formed at an upper portion of each of the active patterns 105. The impurity regions formed at both end portions in an extending direction of each of the active patterns 105 may be electrically connected to the contact plug structure. The impurity region formed at a central portion in the extending direction of each of the active patterns 105 may be electrically connected to the bit line structure 150.

The gate structure 126 may be formed in a first recess positioned at upper portions of the active pattern 105 and the isolation pattern 110. The gate structure 126 may extend lengthwise in the first direction, and a plurality of gate structures 126 may be spaced apart from each other in the second direction. The gate structure 126 may include a gate

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insulation layer 120, a gate electrode 122, and a gate mask 124 sequentially stacked in a vertical direction perpendicular to the upper surface of the substrate 100.

The gate insulation layer 120 may be formed on surfaces of the active pattern 105 and the isolation pattern 110, and the gate electrode 122 may be formed on the gate insulation layer 120 so as to extend in the first direction. Also, the gate mask 124 may cover an upper surface of the gate electrode 122. For example, the gate insulation layer 120 may contact surfaces of the active pattern 105 and the isolation pattern 110, the gate electrode 122 may contact an upper surface of the gate insulation layer 120, and the gate mask 124 may contact the upper surface of the gate electrode 122.

The gate insulation layer 120 may include, e.g., an oxide such as silicon oxide, and the gate electrode 122 may include, e.g., a metal such as tungsten (W), titanium (Ti), or tantalum (Ta), or a metal nitride such as tungsten nitride, titanium nitride, or tantalum nitride. The gate mask 124 may include, e.g., a nitride such as silicon nitride.

The bit line structure 150 may include a lower conductive pattern 140a, a first barrier metal pattern 142a, a first metal pattern 144a, and a first capping pattern 146a sequentially stacked in the vertical direction. In example embodiments, the bit line structure 150 may be formed on the active pattern 105, the isolation pattern 110, and the gate structure 126 so as to extend lengthwise in the second direction.

A portion of a lower surface of the lower conductive pattern 140a may be formed in a first opening positioned at an upper surface of the gate mask 124, the upper surface of the active pattern 105, and the upper surface of the isolation pattern 110 adjacent to the active pattern 105.

The lower conductive pattern 140a may include polysilicon doped with impurities. The first barrier metal pattern 142a may include, e.g., a metal such as titanium (Ti) or tantalum (Ta) and/or a metal nitride such as titanium nitride and tantalum nitride, or the first metal pattern 144a may include, e.g., a metal such as tungsten (W). The first capping pattern 146a may include an insulation material. The first capping pattern 146a may include, e.g., a nitride such as silicon nitride.

The spacer structure 168 may be formed on sidewalls of the bit line structure 150, and thus may extend lengthwise in the second direction. The spacer structure 168 may include a first spacer 160a, a second spacer 164, and a third spacer 166a sequentially stacked on the sidewalls of the bit line structure 150.

The first spacer 160a may contact the sidewalls of the bit line structure 150, the second spacer 164 may cover an outer wall of the first spacer 160a, and the third spacer 166a may cover the outer wall of the second spacer 164. In some embodiments, the second spacer 164 may contact the outer wall of the first spacer 160a, and the third spacer 166a may contact the outer wall of the second spacer 164.

In example embodiments, each of the first and third spacers 160a and 166a may include, e.g., a nitride such as silicon nitride, and the second spacer 164 may include, e.g., an oxide such as silicon oxide. In some example embodiments, each of the first and third spacers 160a and 166a may include, e.g., a nitride such as silicon nitride, and the second spacer 164 may be an air spacer including air.

The sidewalls of the bit line structure 150 positioned in the first opening and a surface of the first opening may be covered by the first spacer 160a. The lower insulation pattern 162 may be formed on the first spacer 160a positioned in the first opening to fill the first opening.

Meanwhile, the first and second insulation patterns 130a and 132a may be sequentially stacked on the active pattern

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105 and the isolation pattern **110** in which the first opening is not formed. A portion of the bit line structure **150** may be formed on the second insulation pattern **132a**.

The first insulation pattern **130a** may include, e.g., an oxide such as silicon oxide. The second insulation pattern **132a** may include, e.g., a nitride such as silicon nitride.

The fence insulation pattern **176** may be formed on the gate structure **126**, and may be disposed between the bit line structures **150**. The fence insulation pattern **176** may include, e.g., a nitride such as silicon nitride.

The contact plug structure may include a first contact plug **190**, a metal silicide pattern **192**, a second contact plug **204**, and a third contact plug **224** sequentially stacked in the vertical direction.

The first contact plug **190** may be disposed at a lower portion of a contact hole defined between the bit line structure **150** and the fence insulation pattern **176**. The first contact plug **190** may be formed on the active pattern **105** and the isolation pattern **110** adjacent thereto. In example embodiments, a lower surface of the first contact plug **190** may be at a lower vertical level than upper surfaces of the isolation pattern **110** and the substrate **100**.

The first contact plug **190** may contact an outer wall of the third spacer **166a**. In example embodiments, an upper surface of the first contact plug **190** may be lower than an upper surface of the first metal pattern **144a** in the bit line structure **150** and higher than an upper surface of the first barrier metal pattern **142a**. The first contact plug **190** may include polysilicon doped with impurities.

The metal silicide pattern **192** may be formed on the upper surface of the first contact plug **190**. In some example embodiments, a lower surface of the metal silicide pattern **192** may contact the upper surface of the first contact plug **190**. The metal silicide pattern **192** may include, e.g., cobalt silicide, nickel silicide, titanium silicide, or the like. In some example embodiments, the metal silicide pattern **192** may not be formed.

The second contact plug **204** may be formed on the metal silicide pattern **192**, and may include, e.g., a metal material. In some example embodiments, a lower surface of the second contact plug **204** may contact an upper surface of the metal silicide pattern **192**. In other example embodiments, when the metal silicide pattern **192** is not formed, the lower surface of the second contact plug **204** may contact the upper surface of the first contact plug **190**. The second contact plug **204** may include a second barrier metal pattern **200b** and a second metal pattern **202b**, and the second barrier metal pattern **200b** may be formed on sidewalls and a bottom of the second metal pattern **202b**, contacting the sidewalls and the bottom surface of the second metal pattern **202b**.

The second barrier metal pattern **200b** may include, e.g., titanium, titanium nitride, tantalum, or tantalum nitride. The second metal pattern **202b** may include, e.g., a metal such as tungsten (W), aluminum (Al), or copper. For example, the second metal pattern **202b** may include tungsten.

A top surface (e.g., an uppermost surface) of the second barrier metal pattern **200b** may be lower than a top surface of the second metal pattern **202b**. For example, the second barrier metal pattern **200b** may not be formed on an uppermost sidewall of the second metal pattern **202b**.

A top surface of the second contact plug **204** may be lower than a top surface of the bit line structure **150**.

An upper recess may be included at a portion of an upper surface of the second contact plug **204**. In example embodiments, the upper surface of the second contact plug **204** on which the upper recess is not formed may have a relatively high height, and the upper surface of the second contact plug

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204 on which the upper recess is formed has a relatively low height. For example, the upper surface of the second contact plug **204** on which the upper recess is formed may have a height that is lower than the height of the second contact plug **204** on which the upper recess is not formed.

In example embodiments, a bottom of the upper recess may be higher than a position of $\frac{1}{3}$ of a total height of the second contact plug **204**. For example, the bottom of the upper recess may be higher than a position of $\frac{1}{2}$ of the total height of the second contact plug **204**.

The second insulating interlayer **230** may be formed on the second contact plug **204**, the fence insulation pattern **176**, and the bit line structure **150**. The second insulating interlayer **230** may contact upper and side surfaces of the second contact plug **204**, the fence insulation pattern **176**, and the bit line structure **150**. The second insulating interlayer **230** may include, e.g., silicon nitride. An upper surface of the second insulating interlayer **230** may be higher than an upper surface of the bit line structure **150**. The upper surface of the second insulating interlayer **230** may be substantially flat. For example, the upper surface of the second insulating interlayer **230** may be planar. Terms such as "same," "equal," "planar," or "coplanar," as used herein when referring to orientation, layout, location, shapes, sizes, amounts, or other measures, do not necessarily mean an exactly identical orientation, layout, location, shape, size, amount, or other measure, but are intended to encompass nearly identical orientation, layout, location, shapes, sizes, amounts, or other measures within acceptable variations that may occur, for example, due to manufacturing processes.

A fourth opening **210** may be an etched portion of the second insulating interlayer **230**, the first capping pattern **146a** of the bit line structure **150**, the spacer structure **168**, and the second contact plug **204**. A lower portion of the fourth opening **210** may correspond to the upper recess.

The fourth opening **210** may include a first sidewall portion exposing the first capping pattern **146a** and the spacer structure **168** and a second sidewall portion exposing an upper portion of the second contact plug **204**. In the fourth opening **210**, the first sidewall portion may have a gentle slope more than the second sidewall portion. For example, a portion of the first sidewall portion may have a slope that is less than 90 degrees with respect to an upper surface of the third metal pattern **222a**, and the second sidewall portion may have a slope that is 90 degrees with respect to the upper surface of the third metal pattern **222a**. In example embodiments, the slope of the first sidewall portion may be in the range of 40 to 65 degrees with respect to the upper surface of the third metal pattern **222a**. Thus, in a portion lower than a top surface of the first capping pattern **146a**, an inner width of the fourth opening **210** may gradually decrease downward. Upper portions of the first capping pattern **146a** and the spacer structure **168** may have an oblique inclination corresponding to the slope of the first sidewall portion of the fourth opening. For example, the upper portions of the first capping pattern **146a** and the spacer structure **168** may have a slope that is greater than 90 degrees with respect to an upper surface of the second insulating interlayer **230**. In example embodiments, the upper portions of the first capping pattern **146a** and the spacer structure **168** may have a slope that is in the range of 115 to 140 degrees with respect to an upper surface of the second insulating interlayer **230**. Accordingly, both upper sidewalls of a structure including the bit line structure **150** and the spacer structure **168** may not have the same slope, in a cross-sectional view.

The third contact plug 224 may be formed at an inner portion of the fourth opening 210. The third contact plug 224 may be formed on the upper recess of the second contact plug 204. The third contact plug 224 may fill the upper recess, and may protrude from an upper portion of the upper recess. The third contact plug 224 may contact the first capping pattern 146a in the bit line structure 150, the surface of the spacer structure 168, and an upper portion of the second contact plug 204. In example embodiments, a lowermost surface of the third contact plug 224 may be higher than a position of $\frac{1}{3}$ of the total height of the second contact plug 204. For example, the lowermost surface of the third contact plug 224 may be higher than a position of $\frac{1}{2}$ of the total height of the second contact plug 204.

Both sidewalls of the third contact plug 224 may have slopes different from each other, in a cross-sectional view. A first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have a gentle slope more than a second sidewall facing the first sidewall of the third contact plug 224. The first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have a gentle slope more than a sidewall of the third contact plug 224 contacting the second contact plug 204. For example, the sidewall of the third contact plug 224 contacting the second contact plug 204 may have a vertical slope with respect to an upper surface of the third contact plug 224, and the first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have an oblique slope. For example, the first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have a slope that is less than 90 degrees with respect to the upper surface of the third metal pattern 222a, and the sidewall of the third contact plug 224 contacting the second contact plug 204 may have a slope that is 90 degrees with respect to the upper surface of the third metal pattern 222a. In example embodiments, the first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have a slope that is in the range of 40 to 65 degrees with respect to the upper surface of the third metal pattern 222a. In the position lower than the top surface of the first capping pattern 146a, a width of the third contact plug 224 may gradually decrease downward. Upper sidewalls of the first capping pattern 146a and the spacer structure 168 contacting the third contact plug 224 may have a gentle slope more than opposite upper sidewalls of the first capping pattern 146a and the spacer structure 168.

Accordingly, a distance d between a sloped sidewall of the third contact plug 224 and the second contact plug 204 adjacent thereto may increase. Therefore, a bridge defect between the third contact plug 224 and a contact plug structure adjacent thereto may be decreased.

The third contact plug 224 may include a metal material. The third contact plug 224 may include a third barrier metal pattern 220a and a third metal pattern 222a, and the third barrier metal pattern 220a may be formed on sidewalls and a bottom of the third metal pattern 222a. For example, the third barrier metal pattern 220a may contact the sidewalls and the bottom surface of the third metal pattern 222a. The third barrier metal pattern 220a may be formed on a surface of the upper recess of the second contact plug 204.

The third barrier metal pattern 220a may include, e.g., titanium, titanium nitride, tantalum, or tantalum nitride. The third metal pattern 222a may include, e.g., a metal such as tungsten (W), aluminum (Al), or copper. For example, the third metal pattern 222a may include tungsten.

A top surface of the third contact plug 224 may be higher than a top surface of the bit line structure 150. A lowermost surface of the third contact plug 224 may be lower than the top surface of the second contact plug 204. An upper surface of the third contact plug 224 may be planar. Upper surfaces of the second insulating interlayer 230 and the third contact plug 224 may be coplanar with each other.

In example embodiments, a plurality of third contact plugs 224 may be spaced apart from each other in each of the first and second directions. The third contact plugs 224 may be disposed to have a honeycomb-type arrangement, such that third contact plugs 224 may be disposed at respective vertices and centers of a regular hexagon, in a plan view. An upper surface of each of the third contact plugs 224 may have a circular shape, an oval shape, or a polygonal shape.

The etch stop layer 232 may be formed on the second insulating interlayer 230 and the third contact plug 224. A lower surface of the etch stop layer 232 may contact upper surfaces of the second insulating interlayer 230 and the third contact plug 224.

The capacitor 240 may pass through the etch stop layer 232, and may contact the third contact plug 224. The capacitor 240 may include a lower electrode 240a, a dielectric layer 240b, and an upper electrode 240c sequentially stacked.

The lower electrode 240a and the upper electrode 240c may include, e.g., doped polysilicon, metal nitride, and/or metal. The dielectric layer 240b may include, e.g., an oxide such as silicon oxide or metal oxide and/or a nitride such as silicon nitride. The metal may include aluminum (Al), zirconium (Zr), or titanium (Ti), hafnium (Hf), or the like.

In the semiconductor device, the second contact plug 204 and the third contact plug 224 may be respectively formed by different processes. In the second contact plug 204, the top surface of the second barrier metal pattern 200b may be lower than the top surface of the second metal pattern 202b. Accordingly, bridge defects between adjacent contact plug structures caused by remaining second barrier metal pattern 200b over the second metal pattern 202b may be decreased.

The sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have the oblique slope. The width of the third contact plug 224 may be gradually decreased downward. Accordingly, the distance between the sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 and the second contact plug 204 adjacent thereto may be increased. Therefore, bridge defects between adjacent contact plug structures may be decreased.

In addition, the third contact plug 224 may maintain a sufficient upper width, so that defects such as a broken upper portion of the third contact plug 224 or increasing of resistance of the third contact plug 224 due to decreasing of the upper width of the third contact plug 224 may be decreased.

FIGS. 4 to 26 are cross-sectional views and plan views illustrating a method of manufacturing a semiconductor device in accordance with example embodiments.

Particularly, FIGS. 5, 7 and 11 are plan views, and FIGS. 4, 6, 8 to 10, and 12 to 26 are cross-sectional views. The cross-sectional views include cross-sections taken along lines A-A' and B-B' of the plan views.

Referring to FIGS. 4 and 5, an active pattern 105 may be formed on a substrate 100. An isolation pattern 110 covering a sidewall of the active pattern 105 may be formed on the substrate.

The active pattern 105 may be formed by removing an upper portion of the substrate 100 to form a trench. The

isolation pattern **110** may be formed by forming an isolation layer on the substrate **100** to fill the trench on the substrate **100**, and then planarizing the isolation layer until an upper surface of the active pattern **105** is exposed. In example embodiments, the planarization process may include a chemical mechanical polishing (CMP) process and/or an etch-back process.

Referring to FIGS. **6** and **7**, an impurity region (not shown) may be formed in the substrate **100** by performing, e.g., an ion implantation process. The active pattern **105** and the isolation pattern **110** are partially etched to form a first recess extending in the first direction.

Thereafter, a gate structure **126** may be formed in the first recess. The gate structure **126** may include a gate insulation layer **120**, a gate electrode **122**, and a gate mask **124** stacked. The gate structure **126** may extend lengthwise in the first direction. The gate structure **126** may include a plurality of gate structures **126**, and the plurality of gate structures **126** may be spaced apart from each other in the second direction.

Referring to FIG. **8**, an insulation layer structure **134**, a first conductive layer **136**, and a first etch mask (not shown) may be sequentially formed on the active pattern **105**, the isolation pattern **110**, and the gate mask **124**. The first conductive layer **136**, the insulation layer structure **134**, and an upper portion of the active pattern may be etched using the first etch mask to form a first opening **138**. The active pattern **105** may be exposed by a bottom of the first opening **138**.

In example embodiments, the insulation layer structure **134** may include first and second insulation layers **130** and **132** sequentially stacked. The first insulation layer **130** may include, e.g., an oxide such as silicon oxide, and the second insulation layer **132** may include, e.g., a nitride such as silicon nitride.

The first conductive layer **136** may include, e.g., polysilicon doped with impurities, and the first etch mask may include, e.g., a nitride such as silicon nitride.

In example embodiments, the first opening **138** may expose a center portion of the upper surface of each of the active patterns **105** extending in the third direction. A plurality of first openings **138** may be arranged in each of the first and second directions.

Thereafter, a second conductive layer **137** may be formed to fill the first opening **138**.

In example embodiments, a preliminary second conductive layer may be formed on the active pattern **105**, the isolation pattern **110**, and the first conductive layer **136** to fill the first opening **138**. An upper portion of the preliminary second conductive layer may be removed by a CMP process and/or an etch-back process. Thus, the second conductive layer **137** may be formed to have an upper surface coplanar with an upper surface of the first conductive layer **136**.

In example embodiments, a plurality of second conductive layers **137** may be formed to be spaced apart from each other in each of the first and second directions. The second conductive layer **137** may include, e.g., polysilicon doped with impurities, and thus the second conductive layer **137** may be merged with the first conductive layer **136**. Hereinafter, a merged layer including the first conductive layer **136** and the second conductive layer **137** is referred to as a lower conductive layer **140**.

Referring to FIG. **9**, a first barrier metal layer **142**, a first metal layer **144**, and a first capping layer **146** may be sequentially formed on the lower conductive layer **140**.

The first barrier metal layer **142** may include, e.g., a metal such as titanium (Ti) or tantalum (Ta) and/or a metal nitride such as titanium nitride or tantalum nitride. The first metal

layer **144** may include, e.g., a metal such as tungsten (W). The first capping layer **146** may include, e.g., a nitride such as silicon nitride.

Referring to FIGS. **10** and **11**, the first capping layer **146** may be patterned to form a first capping pattern **146a**. The first metal layer **144**, the first barrier metal layer **142**, and the lower conductive layer **140** may be sequentially etched using the first capping pattern **146a** as an etching mask.

Thus, a lower conductive pattern **140a**, a first barrier metal pattern **142a**, a first metal pattern **144a**, and the first capping pattern **146a** may be formed on the second insulation layer and the active pattern **105** in the first opening **138**.

A stacked structure including the lower conductive pattern **140a**, the first barrier metal pattern **142a**, the first metal pattern **144a**, and the first capping pattern **146a** may serve as the bit line structure **150**. The bit line structure **150** may extend lengthwise in the second direction. A plurality of bit line structures **150** may be arranged in the first direction.

Referring to FIG. **12**, a first spacer layer **160** may be formed on a surface of the bit line structure **150**, the active pattern **105** and the isolation pattern **110** exposed by the first opening **138**, side surfaces of the first insulation layer **130**, and side and top surfaces of the second insulation layer **132**. The first spacer layer **160** may cover the surface of the bit line structure **150**. A lower insulation pattern **162** may be formed on the first spacer layer **160** to fill the first opening **138**.

The first spacer layer **160** may include, e.g., a nitride such as silicon nitride.

For forming the lower insulation pattern **162**, an insulation layer may be formed on the first spacer layer **160** to fill the first opening **138**, and thereafter an etching process of the insulation layer may be performed. In example embodiments, the etching process may include a wet etching process. In the etching process, the insulation layer except a portion formed in the first opening **138** may be mostly removed to form the lower insulation pattern **162**.

Referring to FIG. **13**, a second spacer layer may be formed on the surface of the first spacer layer **160**, the lower insulation patterns **162** formed in the first opening **138**, and the second insulation layer **132**. The second spacer layer may be anisotropically etched to form a second spacer **164** covering sidewalls of the bit line structure **150**. The second spacer **164** may be formed on the surface of the first spacer layer **160** and the lower insulation pattern **162**.

The second spacer **164** may include, e.g., an oxide such as silicon oxide.

Thereafter, the first spacer layer **160** and the first and second insulation layers **130** and **132** may be etched by a dry etching process using the first capping pattern **146a** and the second spacer **164** as an etch mask. Thus, a first spacer **160a** may be formed to cover the sidewalls of the bit line structure **150**. Also, the first and second insulation layers **130** and **132** are partially etched to form first and second insulation patterns **130a** and **132a**, respectively.

By the above process, the active pattern **105**, the isolation pattern **110**, and the gate mask **124** may be exposed at a space between the bit line structures **150**.

Referring to FIG. **14**, a third spacer layer **166** may be formed on the first spacer **160a**, the second spacer **164**, sidewalls of the first and second insulation patterns **130a** and **132a**, and upper surfaces of the active pattern **105**, the isolation pattern **110**, and the gate mask **124**.

The third spacer layer **166** may include, e.g., a nitride such as silicon nitride.

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After forming the first insulating interlayer **170** on the third spacer layer **166**, the first insulating interlayer **170** may be planarized until an upper surface of the third spacer layer **166** is exposed.

The first insulating interlayer **170** may include, e.g., an oxide such as silicon oxide. The planarization process may include a CMP process and/or an etch-back process.

Referring to FIG. **15**, a second etching mask (not shown) may be formed on the first insulating interlayer **170** and the third spacer layer **166**. The insulating interlayer **170** may be etched by dry etching process using the second etching mask to form a second opening.

In example embodiments, the second etching mask may extend in the first direction, and a plurality of second etching masks may be formed to be spaced apart from each other in the second direction. Each of the second etching masks may not overlap the gate structure **126**, and a space between the second etching masks may overlap the gate structure **126**. Accordingly, the second opening may be formed to overlap the gate structure **126**.

Thereafter, the second etch mask may be removed to expose the upper surface of the first insulating interlayer **170**. A fence insulation pattern **176** may be formed to fill the second opening. The fence insulation pattern **176** may include, e.g., a nitride such as silicon nitride.

Referring to FIG. **16**, the first insulating interlayer **170** may be removed. In example embodiments, the first insulating interlayer **170** may be removed by a wet etching process.

Thereafter, the third spacer layer **166** positioned on spaces between the bit line structures **150** and on the upper surface of the bit line structure **150** may be anisotropically etched to form a third spacer **166a** covering the sidewalls of the bit line structure **150**. A merged structure including the first to third spacers **160a**, **164**, and **166a** is referred to as a spacer structure **168**.

In addition, an upper portion of the active pattern **105** disposed under the third spacer **166a** may be etched by a dry etching to form a third opening **180**. The third opening **180** may be defined by the bit line structure **150** and the fence insulation pattern **176**, and may have an isolated shape. In the dry etching process, an upper portion of the isolation pattern **110** adjacent to an upper portion of the active pattern **105** may also be etched. During the dry etching process, the first to third spacers **160a**, **164**, and **166a** formed on an upper surface of the first capping pattern **146a** are also removed, so that the upper surface of the first capping pattern **146a** may be exposed.

Referring to FIG. **17**, a first contact plug **190** may be formed to fill a lower portion of the third opening **180**.

Particularly, a fourth conductive layer may be formed on the active pattern **105** and the isolation pattern exposed by the third opening **180**, the spacer structure **168**, the first capping pattern **146a**, and the fence insulation pattern. An upper portion of the fourth conductive layer may be removed to form the first contact plug **190**. The upper portion of the fourth conductive layer may be removed by a CMP process and/or an etch-back process.

The first contact plug **190** may include, e.g., polysilicon doped with impurities. In example embodiments, an upper surface of the first contact plug **190** may be lower than an upper surface of the first metal pattern **144a** in the bit line structure **150**.

Thereafter, a metal silicide pattern **192** may be formed on the upper surface of the first contact plug **190**. In example embodiments, an upper surface of the metal silicide pattern **192** may be at a lower level than an upper surface of the first

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metal pattern **144a** in the bit line structure **150**. The metal silicide pattern **192** may include, e.g., cobalt silicide, nickel silicide, titanium silicide, or the like. In some example embodiments, a process for forming the metal silicide pattern **192** may be omitted.

Referring to FIG. **18**, a second barrier metal layer **200** may be conformally formed on the metal silicide pattern **192**, the spacer structure **168**, the first capping pattern **146a**, and the fence insulation pattern **176**. A second metal layer **202** may be formed on the second barrier metal layer **200**.

The second barrier metal layer **200** may include, e.g., titanium, titanium nitride, tantalum, or tantalum nitride. The second metal layer **202** may include, e.g., a metal such as tungsten (W), aluminum (Al), or copper. For example, the second metal layer **202** may include tungsten.

The second metal layer **202** may completely fill a space between the bit line structures, and an upper surface of the second metal layer **202** may be higher than the upper surface of the bit line structure **150**.

Referring to FIG. **19**, the second metal layer **202** and the second barrier metal layer **200** may be planarized until the upper surface of the first capping pattern **146a** in the bit line structure **150** is exposed. The planarization process may include a chemical mechanical polishing process.

When the planarization process is performed, each of the second metal layer **202** and the second barrier metal layer **200** may be separated to form a preliminary second metal pattern **202a** and a preliminary second barrier metal pattern **200a** in the third opening **180**.

Referring to FIG. **20**, upper portions of the preliminary second metal pattern **202a** and the preliminary second barrier metal pattern **200a** may be removed by an etch-back process to form a second metal pattern **202b**. The etch-back process may include a wet etching process.

Thereafter, an upper portion of the preliminary second barrier metal pattern **200a** may be further removed to form a second barrier metal pattern **200b**. The removing process of the preliminary second barrier metal pattern **200a** may include a wet etching process.

Accordingly, a second contact plug **204** including the second barrier metal pattern **200b** and the second metal pattern **202b** may be formed. A top surface of the second barrier metal pattern **200b** may be lower than a top surface of the second metal pattern **202b**.

Atop surface of the second contact plug **204** may be lower than a top surface of the bit line structure **150**. In the second contact plug **204**, the second barrier metal pattern **200b** may cover sidewalls and a bottom of the second metal pattern **202b**, and the second barrier metal pattern **200b** may not be formed on the upper surface of the second metal pattern **202b**.

As described above, the removing process of the upper portion of the preliminary second barrier metal pattern **200a** may be further performed, so that defects (e.g., bridge defects between adjacent contact plug structures) due to remaining the preliminary second barrier metal pattern **200a** over the second metal pattern **202b** may be decreased.

In some example embodiments, the second spacer in the spacer structure **168** may be removed to form an air spacer.

Referring to FIG. **21**, a sacrificial insulation layer **206** may be formed on the spacer structure **168**, the first capping pattern **146a**, the second contact plug **204**, and the fence insulation pattern **176** to cover the bit line structure **150**.

The sacrificial insulation layer **206** may include silicon oxide. The sacrificial insulation layer **206** may include, e.g., a tetraethyl orthosilicate (TEOS) material.

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Referring to FIG. 22, a mask pattern (not shown) may be formed on the sacrificial insulation layer 206. The mask pattern may include exposed portions, and each of exposed portions may face a portion of the bit line structure 150 and a portion of the second contact plug 204.

The sacrificial insulation layer 206 may be anisotropically etched using the mask pattern as an etching mask to expose an upper portion of the second contact plug 204. In the anisotropic etching process, upper portions of the first capping pattern 146a in the bit line structure 150 and the spacer structure 168 may be etched together. However, the upper portions of the first capping pattern 146a and the spacer structure 168 may be etched to have an etch rate lower than that of the sacrificial insulation layer 206. Thus, the upper portions of the first capping pattern 146a and the spacer structure 168 may have a slope that is smaller than opposite upper sidewalls of the first capping pattern 146a and the spacer structure 168.

Thereafter, an exposed upper portion of the second contact plug 204 may be anisotropically etched to form an upper recess. The upper recess may be formed at an upper surface of the second contact plug 204. In the anisotropic etching process, portions of the first capping pattern 146a and the spacer structure 168 may be etched together. However, the first capping pattern 146a and the spacer structure 168 may be etched to have an etch rate lower than that of the sacrificial insulation layer 206. Thus, the upper portions of the first capping pattern 146a and the spacer structure 168 may be etched to have a sidewall slope.

As such, upper portions of the sacrificial insulation layer 206 and the second contact plug 204 may be anisotropically etched to form a fourth opening 210. The sacrificial insulation layer 206, the bit line structure 150, the spacer structure 168, and the second contact plug 204 may be exposed by an inner surface of the fourth opening 210.

A sidewall of the fourth opening 210 exposing the first capping pattern 146a and the spacer structure 168 may have a slope in a direction toward the second contact plug 204. The sidewall of the fourth opening 210 exposing the first capping pattern 146a and the spacer structure 168 may have a gentle slope more than a sidewall of the fourth opening 210 exposing the second contact plug 204. Thus, a distance between the sloped sidewall of the fourth opening 210 and the second contact plug 204 may be increased.

Referring to FIG. 23, a third barrier metal layer 220 may be conformally formed on surfaces of the fourth opening 210 and the sacrificial insulation layer 206. The third barrier metal layer 220 may be formed on the sacrificial insulation layer 206, the bit line structure 150, the spacer structure 168, and the second contact plug 204. A third metal layer 222 may be formed on the third barrier metal layer 220. The third metal layer 222 may be formed to completely fill the fourth opening 210, and an upper surface of the third metal layer 222 may be higher than the top surface of the bit line structure 150.

The third barrier metal layer 220 may include, e.g., titanium, titanium nitride, tantalum, or tantalum nitride. The third metal layer 222 may include, e.g., a metal such as tungsten (W), aluminum (Al), or copper. For example, the third metal layer 222 may include tungsten.

Referring to FIG. 24, the third metal layer 222 and the third barrier metal layer 220 formed on the sacrificial insulation layer 206 may be removed. Subsequently, the third metal layer 222 and the third barrier metal layer 220 formed in an upper portion of the fourth opening 210 may

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be removed. The removing process may include an etch-back process using a dry etching process.

When the etch-back process is performed, each of the third metal layer 222 and the third barrier metal layer 220 may be separated by the fourth opening 210 to form a third contact plug 224 including a third metal pattern 222a and a third barrier metal in the fourth opening 210.

An upper surface of the third contact plug 224 may be lower than an upper surface of the sacrificial insulation layer 206. The upper surface of the third contact plug 224 may be higher than the top surface of the bit line structure 150.

The third contact plug 224 may be formed on a portion of the second contact plug 204. The third contact plug 224 may at least fill the upper recess of the second contact plug 204. The third barrier metal pattern 220a may be formed on a surface of the upper recess of the second contact plug 204.

In example embodiments, a plurality of third contact plugs 224 may be formed to be spaced apart from each other in each of the first and second directions. In a plan view, the plurality of third contact plugs 224 may be arranged in a honeycomb structure. Also, each of the third contact plugs 224 may have a polygonal shape, a circular shape, or an oval shape, in the plan view.

A contact structure may include the first contact plug 190, the metal silicide pattern 192, the second contact plug 204, and the third contact plug 224 sequentially stacked. The upper surface of the third contact plug 224 may serve as a landing pad pattern for contacting a lower electrode in the capacitor.

The third contact plug 224 may contact the upper recess of the second contact plug 204. A lowermost surface of the third contact plug 224 may be lower than the top surface of the second contact plug 204.

The top surface of the second contact plug 204 may be lower than the top surface of the bit line structure 150. Also, the top surface of the second barrier metal pattern 200b included in the second contact plug 204 may be lower than the top surface of the second metal pattern 202b. Thus, a defect in which adjacent contact structures may be electrically connected to each other by the second barrier metal pattern 200b may be decreased.

A first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 may have a gentle slope more than a second sidewall of the third contact plug 224 contacting the second contact plug 204. Thus, a distance between the first sidewall of the third contact plug 224 contacting the first capping pattern 146a and the spacer structure 168 and the second contact plug 204 adjacent thereto may be increased. Thus, a bridge defect between adjacent contact plug structures may be decreased.

Referring to FIG. 25, the sacrificial insulation layer 206 may be removed. The removing process of the sacrificial insulation layer 206 may include a wet etching process.

A second insulating interlayer 230 may be formed to cover the spacer structure 168, the first capping pattern 146a, the second and third contact plugs 204 and 224, and the fence insulation pattern 176 exposed by removing the sacrificial insulation layer 206. The second insulating interlayer 230 may be formed to sufficiently fill a gap between the third contact plugs 224. Thereafter, the second insulating interlayer 230 may be planarized until the upper surface of the third contact plug 224 may be exposed.

The second insulating interlayer 230 may include, e.g., silicon nitride. The second insulating interlayer 230 and the fence insulation pattern 176 include the same material, so that the second insulating interlayer 230 and the fence insulation pattern 176 may be merged into one layer.

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Referring to FIG. 26, an etch stop layer 232 may be formed on the second insulating interlayer 230 and the third contact plug 224. A capacitor 240 may be formed on the upper surface of the third contact plug 224 through the etch stop layer 232.

The capacitor 240 may include a lower electrode 240a, a dielectric layer 240b, and an upper electrode 240c stacked.

Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

What is claimed is:

1. A semiconductor device, comprising:
bit line structures on a substrate;
a contact plug structure on the substrate between the bit line structures, the contact plug structure including a first contact plug, a second contact plug, and a third contact plug sequentially stacked; and
a capacitor electrically connected to the contact plug structure,
wherein an upper surface of the second contact plug includes an upper recess,
wherein the third contact plug fills the upper recess and protrudes above the upper recess, and an upper surface of the third contact plug is higher than a top surface of the bit line structures, and
wherein the second contact plug includes a second barrier metal pattern comprising a first material and a second metal pattern comprising a second material different from the first material, the second barrier metal pattern being on and contacting sidewalls and a bottom of the second metal pattern.
2. The semiconductor device of claim 1,
wherein an upper portion of each of the bit line structures includes a capping pattern including an insulation material, and
wherein a first sidewall of the third contact plug contacts a sidewall of the capping pattern in the bit line structure.
3. The semiconductor device of claim 2, wherein the first sidewall of the third contact plug contacting the capping pattern has a smaller slope than a second sidewall of the third contact plug contacting the second contact plug with respect to an upper surface of the third contact plug.
4. The semiconductor device of claim 2, wherein in a portion lower than a top surface of the capping pattern, a width of the third contact plug is gradually decreased downward.
5. The semiconductor device of claim 1, wherein a top surface of the second contact plug is lower than the top surface of the bit line structures.
6. The semiconductor device of claim 1, wherein the first contact plug includes polysilicon, and the second contact plug and the third contact plug include a metal.
7. The semiconductor device of claim 1, wherein a top surface of the second barrier metal pattern is lower than a top surface of the second metal pattern.

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8. The semiconductor device of claim 1, wherein the third contact plug includes a third barrier metal pattern and a third metal pattern, and the third barrier metal pattern is formed on sidewalls and a bottom of the third metal pattern.

9. The semiconductor device of claim 8, wherein the third barrier metal pattern is formed on a surface of the upper recess of the second contact plug.

10. The semiconductor device of claim 1, further comprising a metal silicide pattern between the first contact plug and the second contact plug.

11. The semiconductor device of claim 1, further comprising a spacer structure on sidewalls of the bit line structures.

12. The semiconductor device of claim 11, wherein the spacer structure comprises:

- a first spacer covering the sidewalls of the bit line structures;
- a second spacer covering an outer wall of the first spacer; and
- a third spacer covering an outer wall of the second spacer.

13. The semiconductor device of claim 11,
wherein the third contact plug contacts an upper portion of the bit line structures and an upper portion of the spacer structure adjacent thereto, and
wherein upper sidewalls of the bit line structures and the spacer structure contacting the third contact plug has a smaller slope than opposite upper sidewalls of the bit line structures and the spacer structure.

14. A semiconductor device, comprising:
bit line structures on a substrate;
a contact plug structure on the substrate between the bit line structures, the contact plug structure including a first contact plug including polysilicon, a second contact plug including a metal on the first contact plug, and a third contact plug including a metal on the second contact plug sequentially stacked; and
a capacitor electrically connected to the contact plug structure, wherein a top surface of the second contact plug is lower than a top surface of the bit line structures, wherein an upper surface of the third contact plug is higher than the top surface of the bit line structures, and a lowermost surface of the third contact plug is lower than the top surface of the second contact plug,
wherein a first sidewall of the third contact plug contacts a sidewall of a first bit line structure of the bit line structure,
wherein the second contact plug includes a second barrier metal pattern comprising a first material and a second metal pattern comprising a second material different from the first material, the second barrier metal pattern being on and contacting sidewalls and a bottom of the second metal pattern, and
wherein the third contact plug contacts the second barrier metal pattern and the second metal pattern.

15. The semiconductor device of claim 14, wherein the first sidewall of the third contact plug contacting the sidewall of the first bit line structure has a smaller slope than a second sidewall of the third contact plug contacting the second contact plug.

16. The semiconductor device of claim 14, further comprising a spacer structure on sidewalls of the first bit line structure.

17. The semiconductor device of claim 16,
wherein the third contact plug contacts an upper portion of the first bit line structure and an upper portion of the spacer structure adjacent thereto, and

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wherein upper sidewalls of the first bit line structure and the spacer structure contacting the third contact plug has a smaller slope than opposite upper sidewalls of the first bit line structure and the spacer structure.

18. The semiconductor device of claim **14**,

wherein a top surface of the second barrier metal pattern is lower than a top surface of the second metal pattern.

19. A semiconductor device, comprising:

a substrate including an active pattern defined by an isolation pattern;

gate structures buried in the active pattern and the isolation pattern of the substrate;

bit line structures on the substrate;

a contact plug structure on the active pattern between the bit line structures, the contact plug structure including a first contact plug including polysilicon, a second contact plug including a metal on the first contact plug, and a third contact plug including a metal on the second contact plug sequentially stacked; and

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a capacitor electrically connected to the contact plug structure,

wherein the second contact plug includes a second barrier metal pattern comprising a first material and a second metal pattern comprising a second material different from the first material, the second barrier metal pattern being on and contacting sidewalls and a bottom of the second metal pattern,

wherein the third contact plug contacts a sidewall of a first bit line structure of the bit line structures and upper surfaces of the second barrier metal pattern and the second metal pattern of the second contact plug, and wherein both sidewalls of the third contact plug have slopes different from each other, in a cross-sectional view.

20. The semiconductor device of claim **1**, wherein the third contact plug contacts the second barrier metal pattern and the second metal pattern.

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